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Department of Mechanical, Materials and Manufacturing Engineering

OPTIMISATION OF THE FLUIDIC SYSTEM FOR VITA EXPERIMENT

BEng PROJECT FINAL DISSERTATION

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SUMMARY

This report introduces a bachelor’s individual project as part of the broader visualising in-space TXTL (Cell-free transcription-translation) astropharmaceuticals (VITA) experiment, which aims to demonstrate the feasibility of producing recombinant proteins in a microgravity environment aboard the international space station (ISS) using their international commercial experiment (ICE) cube facilities. Success in the VITA mission holds the potential to transform pharmaceutical manufacturing in space, reducing reliance on external supplies and increasing self-sufficiency for interplanetary missions.

The primary focus for this individual project is the optimisation of the fluidics system integral to the VITA experiment. This system, based on an elastomeric gasket mechanism acting as a membrane, represents an unconventional approach within space systems, designed for the controlled transfer of the S30A rehydration solution during key experiment phases on the ISS. The optimisation challenge was centred around determining the optimal hole sizes in the gasket and reservoir layers to allow for efficient fluid transfer while preventing leakage.

The project's objectives included theoretical modelling, prototype manufacturing, empirical testing and environmental simulation. Theoretical models were constructed to estimate the optimal system parameters, resulting in a range of feasible hole sizes that prioritised minimal force required for fluid transfer. Furthermore, the models inferred the frictional forces being a significant contributor to motor force, prompting a redesign of the plunger mechanism. The introduction of an O-ring plunger design resulted in a notable reduction in frictional forces, enhancing the system’s efficiency. Subsequent iterations and testing of the fluidic system addressed and resolved initial design issues, further refining the system's operational reliability, leading to the selection of a 4.5 mm reservoir diameter paired with a 1.27 mm gasket hole as the optimal configuration, based on its leakage prevention and efficient fluid transfer capabilities.

The project's outcomes confirmed the proposed design's adherence to the theoretical model’s predictions. The finalised configuration showed no leakage and high efficiency fluid transfer, whilst meeting all of the set requirements, providing an optimal soluiton for the VITA experiment.

Future work will focus on environmental testing to validate the fluidic system's durability against launch-induced vibrations. Additionally, further refinement of the system is anticipated, with the integration of heat-sealing techniques and exploration of failsafe design enhancements to mitigate potential failure modes and increase the system’s reliability.

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NOMENCLATURE

Symbol	**Quantity**	**Unit**
F	force	N
P	pressure	Pa
V	velocity	m/s
r	radius	m
d	diameter	m
A	area	m^2
γ	surface tension	N/m
L	length	m
ρ	density	kg/m^3
x	displacement	m
μ	friction coefficient	
k	spring coefficient	

Subscripts/superscripts	**Meaning**
gasket	gasket
reservoir	reservoir
SurfaceTension	surface tension
motor	motor
friction	friction
dynamic	dynamic

ACRONYMS

Acronym	**Meaning**
ESA	European Space Agency
ISS	International Space Station
ICE	International Commercial Experiment
VITA	Visualising In-space TXTL Astropharmaceuticals
SU	Science Unit
FEA	Finite Element Analysis

1. INTRODUCTION

1.1. BACKGROUND AND CONTEXT

The European Space Agency (ESA) offers an educational program known as Orbit Your Thesis, which provides a platform for university student teams to test self-developed experiments in a microgravity environment [1]. The student-devised experiments chosen are launched to the international space station (ISS), and utilise the International Commercial Experiment (ICE) cube facility on board the Columbus module to provide the required experimental conditions [1].

In 2022, the Visualising In-space TXTL (Cell-free transcription-translation) Astropharmaceuticals (VITA) experiment from the University of Nottingham was selected for the Orbit Your Thesis program [2]. The interdisciplinary student-led experiment is designed to explore the feasibility of producing on-demand recombinant proteins from cell extracts upon the addition of DNA and other reagents within a microgravity environment [2]. The experiment's success would be immensely valuable, eliminating the need to keep cells alive, and owing to its flexibility would be able to produce any protein with any DNA input, allowing for personalised medical attention for long-duration space flight [2]. Ultimately this method could revolutionise pharmaceutical manufacturing in space as well as in other extreme environments, potentially facilitating and extending interplanetary missions by reducing external supply dependency and enhancing self-sufficiency [2] The provided concept of operations (CONOPS) figure below depicts the experiments timeline of events.

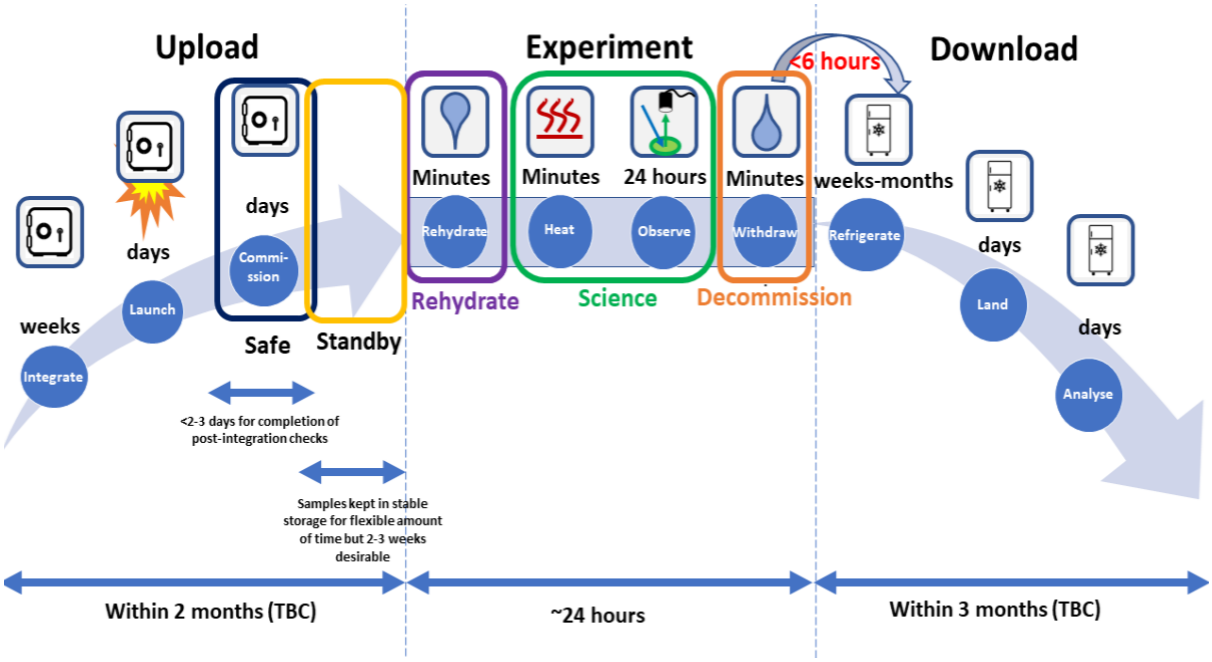

Figure 1: Diagram of the VITA Concept of Operations and Modes of Operations [3].

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Initially, the experimental payload is launched to the ISS within a controlled, refrigerated environment, ensuring the integrity of the biological samples. Once aboard the ISS, integration into the ICE cubes system takes place, preparing the samples for the experimental phase. Over an approximate 24-hour period, the experiment engages in a multi-phase process beginning with the rehydration of cell extract samples using the S30A solution. Observations are recorded to detect indicators of success. Once the experiment is completed, it is decommissioned, at which point the rehydrated samples are safely secured for preservation. Subsequently, the samples are refrigerated to maintain their viability until their return to Earth, where detailed analysis will be conducted.

The rehydration and decommission stages are key to this report as they utilise the fluidic system which is the focus of this individual project optimisation. These processes take place within each science unit (SU), found inside the payload as seen in the figure below.

Figure 2: The four SU modules each containing 4 biological sample wells.

Each SU contains 4 different sample types, meaning each SU is an independent repeat of the experiment, acting as a layer of precaution in case of fluidic system failure in one SU. This greatly reduces the risk of total experiment failure. The fluidic system within each SU needs to store the S30A solution during the upload phase, deliver the necessary 50μL of solution to each sample during the rehydration phase and retrieve the solution with the cell extracts during the decommission phase without any unintended leakages. Figure 4 below gives a detailed exploded view of an individual SU showing each layer involved.

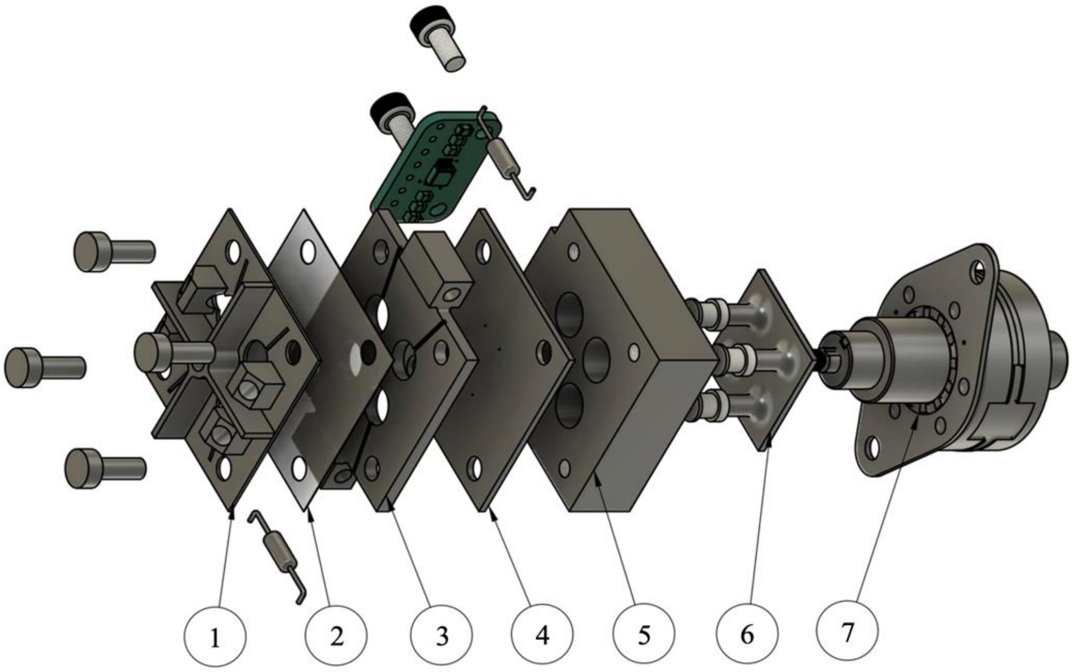

Figure 3: Exploded view of an SU. 1: LED Holder/Shroud layer, 2: Sample Seal layer, 3: Sample Well layer, 4: Pre-punched Gasket layer, 5: Reservoir layer, 6: Plunger layer, 7: Stepper Motor.

The system consists of several layers, including a reservoir where the S30A solution is initially stored, and a sample well layer where the cell extracts are initially stored. The actuation of the system is driven by a stepper motor, which forces the plungers through the reservoir layer holes with the necessary pressure to transfer the solution through the pre-punched gasket layer holes to the sample well layer. This fluid transfer process is further described in the cross section figure below.

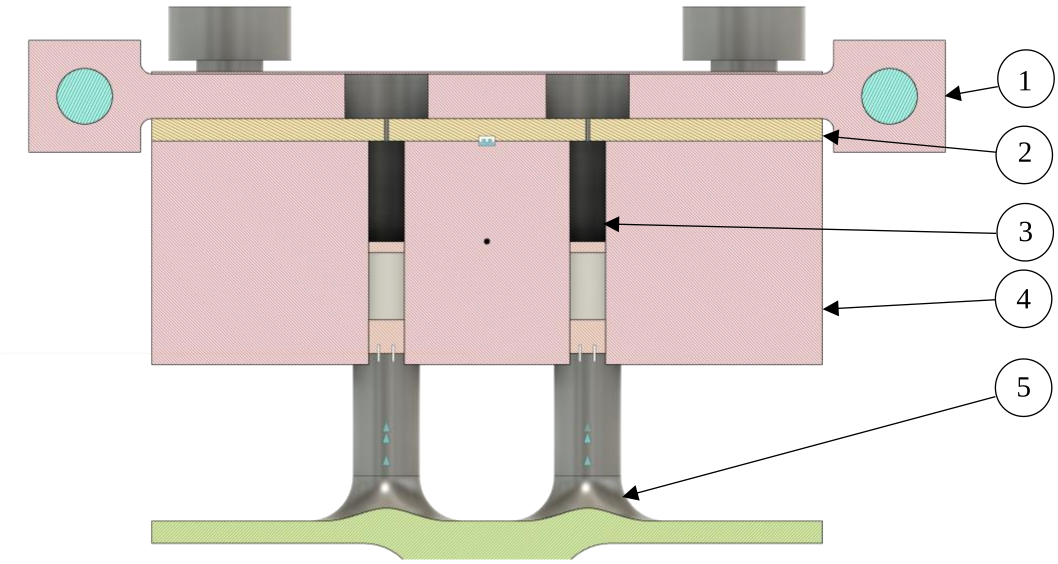

Figure 4: Cross section view of an SU Fluidic System. 1: Sample Well layer, 2: Pre-punched Gasket layer, 3: S30A rehydration solution, 4: Reservoir layer, 5: Plunger layer 2x plungers.

The use of the elastomeric gasket layer for fluid transfer in microgravity represents an innovation within the context of space systems and a departure from traditional valve-based methods. It is designed to allow passage of the rehydration solution only when the plungers apply sufficient pressure. This element of the design is responsible for maintaining the separation between the rehydration solution and the biological samples, ensuring the experiment's integrity.

The project's optimisation challenge lies in ensuring the fluidic system's intended functionality while minimising the energy consumption required for the motor's operation given the limited and therefore budgeted power available in the ICE cube facility. The primary objective of this optimisation is to determine the ideal hole sizes in both the gasket and reservoir layers that would require the least force for successful fluid transfer, without compromising the system's integrity by unintended leakage. Should the holes be of an inadequate size, the buffer solution could leak, particularly during the turbulent launch conditions, to the samples prematurely and jeopardise the experiments success.

For clarity, within this report the terms “fluid transfer” and “leakage” will be used to refer, respectively, to situations where (i) the fluid goes from the reservoir through the gasket holes and to the sample wells upon actuation and (ii) the fluid leaves the reservoir in any non-optimal conditions for example through the sides of the gasket layer or through the bottom of the plungers.

1.2. AIM AND OBJECTIVES

The overall aim of this individual project is to develop a design for the fluidic system in the VITA experiment that can store the S30A solution, perform fluid transfer of the correct volume at the correct times whilst minimising stepper motor force needed for fluid transfer and prevent inadvertent leakages of solution. This design should be evaluated for worst case leakage scenarios occurring from vibrations throughout the rocket taking off and should be able to function effectively in microgravity conditions. Extensive testing must be conducted with the final configuration to ensure proper functionality. A list of objectives which illustrate the approach followed to achieve the project aim can be seen below:

Objectives:

- Create a virtual/theoretical model of the fluidic system.
- Obtain a range of possible gasket and reservoir hole sizes from the model, those of which utilise minimal motor force throughout fluid transfer whilst not allowing leakage to occur.
- Manufacture physical prototypes of the fluidic systems within the acquired range.
- Test the physical prototype fluidic systems to obtain the real world optimal solution.
- Validate the physical prototype through environmental tests.

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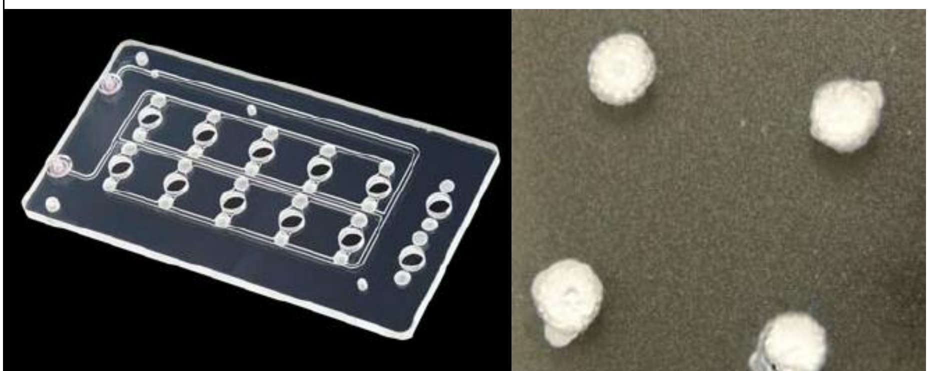
2. LITERATURE REVIEW

2.1. SOLUTIONS FOR FLUIDIC SYSTEM

The fluidic transfer method selected by the VITA team, involving a pre-punched gasket layer, is a novel solution within the use of space systems. The rationale for adopting this specific solution underwent extensive scrutiny, with a focus on similar fluidic configurations and their comparative feasibility against established alternatives. A review of fluidic transfer methods previously and currently utilised in aerospace missions was conducted to determine the gasket layer's suitability for the intended application. Three solutions were considered by the VITA team, including (i) the integration of a filter membrane, (ii) the use of a syringe/plunger system or an air pump to generate negative pressure, and (iii) the incorporation of a pre-punched elastomeric gasket. Subsequent research for this project included investigation into the use of solenoid valves which are a standard in small-scale satellite technologies such as CubeSats, due to their precision and reliability in fluid control. The different options have been outlined with their respective evaluations in the table below:

Table 1: Comparison of different Fluidic System Solutions.

Fluidic System Solution	**Description and Analysis**
Filter Membrane	Filter membranes were pertinent in the Genesat and O/Oreos missions where the samples required transfer from the well at a specific point within the mission [4] [5] similar to the VITA mission. The membranes used as seen in figure 5 can be compact, perfect for the VITA size requirements. These missions were studying E. Coli, Halorubrum chaoviatoris, and Bacillus subtilis which are of a larger size compared to the samples produced in the VITA experiment [3] [4] [5]. Consequently, filter membranes would not serve as a viable option given the risk of sample diffusion back through the membrane.  Figure 5: Left: GeneSat-1 Fluidic Card containing membranes at the inlet/outlet of each sample well, Right: Filter Membranes [6].

Plunger/Air Pump	This approach entails using either a syringe/plunger system or an air pump to generate a negative pressure, drawing the fluid from a closed static reservoir to the samples. This solution would solve displacing the existing air within the sample wells. During testing with the setup seen in figure 6 below, this method exhibited “erratic and unpredictable” behaviour caused by differences in the compressibility of air compared to the S30A solution making it unviable as an option [3].  Figure 6: Left: Tubing section representing a sample well, Right: Experimental setup, with left syringe being the reservoir for S30A and right syringe representing a generic air pump [3].
Pre-punched Elastomeric Gasket	Rubber stoppers, adopted in the pharmaceutical industry for sealing vaccine vials, are able to maintain airtight seals in vials whilst preventing leakages and spills. They also have the ability to enable the controlled extraction of the fluid by applying external pressure, typically via a needle. Once the pressure is removed, the rubber stopper reverts from a deformed state to its original state, once again maintaining a seal. It is capable of this through the materials hyperelastic properties characterised by a “non-linear stress-strain relation” [7] allowing it to undergo substantial deformation and still revert to its original shape, due to all of the inputted “energy being recoverable” [7]. For this reason a similar rubber stopper like design is highly desirable for a project such as this one. In light of the viability in the pre-punched elastomeric gasket design, the design was tested to access its suitability as illustrated in Figure 7, where a silicone vial stopper was positioned between an eppendorf (with a top hole for air passage) and a syringe. The stopper had a pre-punched hole and was securely sealed. Multiple trials were performed, and the outcomes affirmed “that rubber gaskets could successfully open when pressure is applied during the experiment to allow S30A through to rehydrate the samples” and that “they would reseal to prevent sample diffusion after pressure is equalised or removed” [3]. In view of this successful proof of

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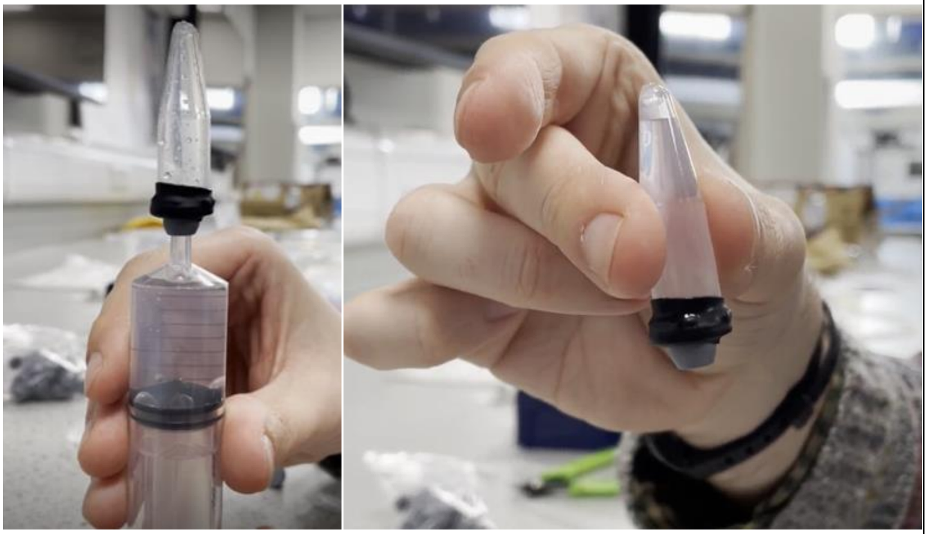
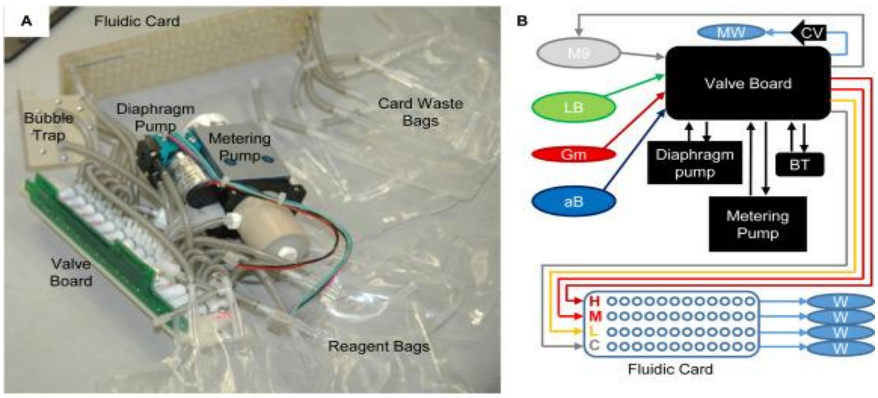
	concept the decision was made to use a gasket composed of a silicon rubber material, incorporating a pre-punctured hole similar to that of the vial stoppers used in the validated test.  Figure 7: Left: Vial Stopper connected to eppendorf on one end and syringe on other using heat seal, Right: Vial Stopper with hole connected to eppendorf demonstrating sealing capability [3].
Solenoid Valve	Solenoid valves serve as a tried and true method widely used in space fluidic systems given their reliability and durability. They have the capability to open or close based on the electric current applied to the armature [8] thereby regulating fluid flow. They were used on the EcAMSat system which had similar fluidic system requirements to the VITA system in terms of fluidic delivery system's pumps and valves for reagent exchanges [9]. Despite the benefits of this approach, further analysis demonstrated that their adoption would be challenging given the space limitations within the experiment payload and would require extensive remodelling and time which was a luxury not available at this stage of the project. Additionally, they would introduce more complexities in terms of both manufacturing and electronics.  Figure 8: EcAMSat Fluidic System controlled by Solenoid Valve PC Board.

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2.2. SPACE ENVIRONMENT CONSIDERATIONS

As the project is partially subject to a space environment, it is important to take into consideration the unique conditions and challenges that arise in such an extreme setting. The system will be exposed to several factors that may affect its performance, including radiation, microgravity and pressure/temperature fluctuations, the effects of which are detailed below:

- Radiation affects the elastomeric properties of silicone by "increasing the modulus of elasticity linearly" producing more "rigid rubbers" [10]. However, this significant change in material properties typically occurs at a dose of around 100 megaRads which far exceeds the "150 to 200 milliRads per day" experienced on the ISS [11], indicating the likely negligible impact of cosmic radiation on the performance of the elastomer material.

- Microgravity is unlikely to exert substantial stress on the self-sealing gasket system and will be considered negligible given the "small gravity field effects ($10^{-6} – 10^{-9}$ m/s^2)" [12].

- The pressure environment within the experiment payload is hermetically sealed at 1 atm however if depressurisation were to occur, the fluidic system components could exhibit outgassing. This could have multiple detrimental effects, including "contamination of surfaces, loss of dimensional stability and changes in material properties" [12]. Therefore pressurisation stability is crucial to ensure the gaskets integrity throughout the experiment.

- Temperature variations are anticipated, with fridge conditions at around 3 degrees Celsius and heating experimental conditions at around 30 degrees Celsius. Such variations in temperature are relatively tame and shouldn't provoke any elastomer changes in "internal and bonding strength" or "cause it to crack, fall off and age" [12].

2.3. NASA AND ECSS STANDARDS

Within the ISS, it is fundamental to account for the NASA and European Cooperation for Space Standardization (ECSS) standards and understand how they can limit the project. A detailed examination of these standards has indicated that they pose no significant barriers toward the fluidic system for the project in question. It does not contribute substantially to thermal load or acoustic noise. Moreover, the system manages a small quantity of biologically inert, non-toxic liquid within a hermetically sealed payload. This design complies with the containment and safety regulations as outlined in the relevant standards [13]. Specific standards and their compliance are listed below:

- **Flammability**: The materials used, specifically silicone, adhere to the flammability criteria as per the specifications of NASA-STD-6001 [14] and ECSS-Q-ST-70-21C [15].

- **Offgassing**: Regarding the concern of offgassing from the pressurized fluidic system, an evaluation will be performed to confirm that any volatile polymeric products are within the limits of NASA-STD-6001B §7.7.3 [16], SSP 30233 [17] and/or ECSS-Q-ST-70-29C [18].

- **Noise**: The noise generated by the motor actuating the plunger will be verified according to the part supplier and if necessary, will be tested in an acoustic noise test in order to fulfil the requirements set forth in §3.12.3.3 of COL-RIBRE-SPE-0164 [19].

3. METHODOLOGY

3.1. MISSION AND SYSTEM REQUIREMENTS

Building upon prior and concurrent VITA experiment development has led to several design considerations which must be taken into account to align with its main objectives. The mission and system requirements pertaining to the design of the fluidic system are listed below [3]:

Mission Requirements:

1. Safely storing the S30A buffer rehydration solution during launch and the in-orbit regime, preventing any leakage to the sample areas or beyond the designated reservoir region.
2. Transferring 40μL of S30A buffer rehydration solution to the sample wells at the appropriate time during the rehyradration phase.
3. Retrieving and storing the samples and S30A rehydration solution during the decomission phase, allowing for safe transport back to Earth for further analysis.

Fluidic System Requirements:

1. Pre-punched gasket holes must be precisely positioned in-line with air evacuation channels to allow for adequate rehydration of samples in microgravity.
2. Reservoir holes must fit within the designated 10 x 10 mm area.
3. Stepper motor force must use minimum required energy for fluid transfer.
4. Layer sealing techniques must not use chemical adhesives, given the risk of cell extract contaminatation, potentially compromising the experiment integrity.

3.2. THEORETICAL FLUIDIC SYSTEM MODEL

Developing a theoretical model of the fluidic system allowed for a more efficient path to fulfilling the project requirements, as opposed to a brute-force approach that would involve extensive physical testing across a wide array of reservoir and gasket hole sizes.

The initial proposal for an optimisation strategy entailed a coupled analysis of the fluidic system within the Ansys workbench [20], utilising both Ansys Mechanical and Ansys Fluent. The aim was to iterate through various gasket and reservoir hole sizes as well as plunger pressures, to ascertain the optimal system dimensions for achieving minimum plunger pressure (therefore motor force), and ensuring fluid transfer without leakage.

Despite soliciting external expertise, the complexity in modelling the interactions between the fluid mesh and the gasket proved challenging. The difficulty mainly stemmed from the need to simulate the dynamic hyperelastic behaviour of the silicon rubber gasket in a coupled environment, a non-trivial task requiring a sophisticated dynamic mesh, which would require extensive research and given the time restraints became unfeasible to complete. Therefore the approach was modified to only include a simplified gasket finite element analysis (FEA) within ABAQUS [21].

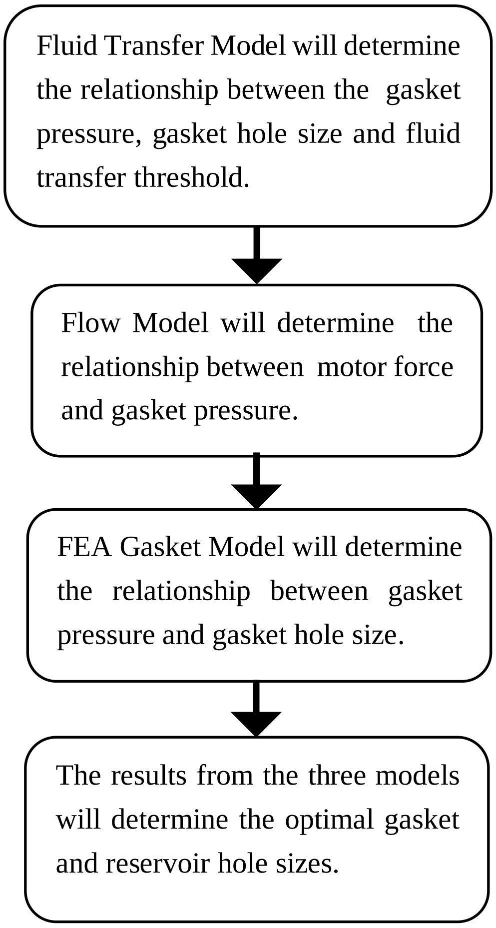

Figure 9: Block diagram of the methodology for optimisation.

Subsequently, a numerical approach (see block diagram to above) partially incorporating an FEA of the rubber silicon gasket was opted for instead, relying on certain assumptions but still allowing for an estimation of optimal fluidic system parameters and force needed for leakage under varying reservoir and gasket hole sizes.

3.2.1 FLUID TRANSFER MODEL

The fluid transfer model is designed to define the relationship between the pressure applied to the gasket hole and the resulting fluid transfer, dependant on gasket hole size. The gasket hole pressure will then be linked to the motor force in the following section. Within the VITA experimental context demonstrated the fluid does not simply transfer through the gasket hole due to the surface tension surrounding the meniscus of the fluid attached to the walls of the hole [22]. For fluid transfer to occur, the applied pressure on the fluid meniscus must exceed the counteracting force exerted by the surface tension. This can be expressed through the force balance equation, associated with fluid transfer threshold, below [23]:

$$F_{gasket} = F_{SurfaceTension} \qquad \text{Equation 1}$$

The force caused by pressure can be expressed as the product of the applied pressure to the meniscus (P_{gasket}) in Pa and the contact area (A_{gasket}) in m^2. Similarly, the force caused by surface tension can be expressed as the product of the surface tension (γ) in N/m and the gasket hole contact length (L) in m.

$$P_{gasket}A_{gasket} = \gamma L \qquad \text{Equation 2}$$

Considering that the contact length around the gasket hole is equivalent to the hole's perimeter, and the area is defined by the hole's cross-sectional area, the equation can be reformulated as follows:

$$P_{gasket} \times \pi r_{gasket}^2 = \gamma \times 2\pi r_{gasket} \qquad \text{Equation 3}$$

The relationship between the pressure applied to the meniscus for fluid transfer to occur and the gasket hole size can be seen below.

$$P_{gasket} = \frac{2\gamma}{r_{gasket}} \qquad \text{Equation 4}$$

Equation 4 allows for the calculation of the hole gasket pressure necessary for water transfer. The applied pressure at the gasket hole must be linked to the pressure at the reservoir hole as seen in the diagram below, to determine its correlation to the force exerted by the motor.

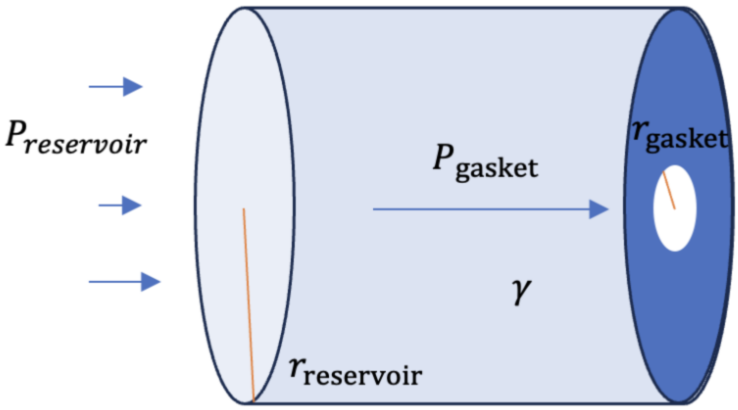

Figure 10: Fluid Transfer Model.

The following Flow Model section will establish a relationship between the motor force and the resultant gasket pressure applied by the plunger to ensure coherent and controlled fluid movement.

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3.2.2 FLOW MODEL

The flow model utilises the Bernoulli's principle in pipe flow [24] to determine the relationship between motor force (F_{motor}) and gasket pressure (P_{gasket}). The motor force can be represented as the pressure applied on the fluid from the plunger heads ($P_{reservoir}$) and the area of the plunger heads which corresponds to the reservoir area ($A_{reservoir}$). Both the pressure and the area will be in terms of the reservoir which corresponds to the entrance of the reservoir as seen below.

$$F_{motor} = P_{reservoir}A_{reservoir} \qquad \text{Equation 5}$$

This model approximates the fluidic system as a tapered pipe, depicted in figure 11. This simplification does not account for the non-laminar nature of the flow, induced by the motor programmed to push the fluid in steps, or for the fluids viscosity, both of which deviate from the Bernoulli principle assumptions.

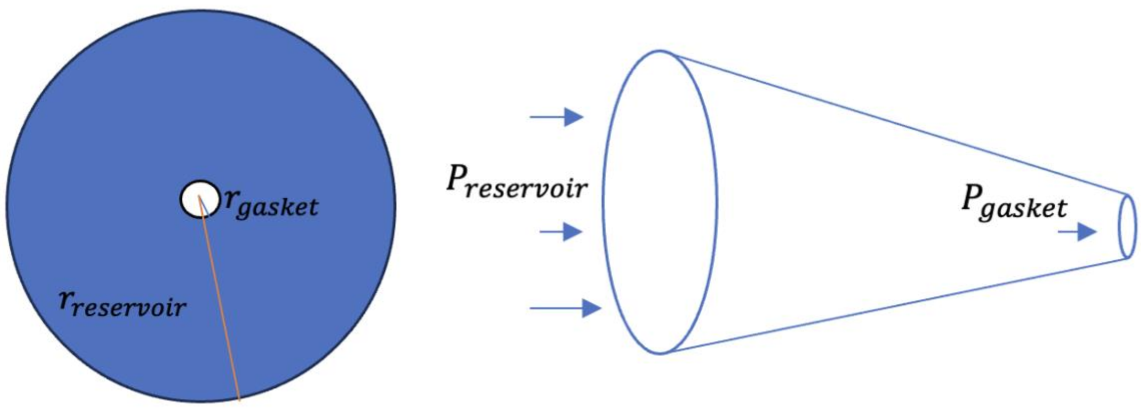

Figure 11: Diagram showing pipe model for hole leakage parameters. Right: Front View, Left: Side View.

In utilizing Bernoulli's equation, the plunger pressure ($P_{reservoir}$) can be linked to the gasket pressure (P_{gasket}), whilst simplifying the terms of potential energy on account of equal hole centre heights:

$$P_{reservoir} + \frac{1}{2}\rho V_{reservoir}^2 = P_{gasket} + \frac{1}{2}\rho V_{gasket}^2 \qquad \text{Equation 6}$$

After utilising the continuity equation to maintain mass conservation, the radii of both the reservoir and gasket holes can be related to fluid flow velocities. Substituting for V_{gasket} from the continuity equation into the simplified Bernoulli equation yields the following:

$$P_{reservoir} - P_{gasket} = \frac{1}{2}\rho\left(V_{reservoir}\left(\frac{r_{reservoir}^2}{r_{gasket}^2} - 1\right)\right)^2 \qquad \text{Equation 7}$$

[redacted]

Next the Fluid Transfer Model Equation 4 can be substituted to express the pressure terms in relation to the fluid transfer threshold:

$$P_{reservoir} - \frac{2\gamma}{r_{gasket}} = \frac{1}{2}\rho\left(V_{reservoir}\left(\frac{r^2_{reservoir}}{r^2_{gasket}} - 1\right)\right)^2 \qquad \textit{Equation 8}$$

Finally when restructured for F_{motor} the desired relationship can be seen below:

$$F_{motor} = \left(\frac{1}{2}\rho\left(V_{reservoir}\left(\frac{r^2_{reservoir}}{r^2_{gasket}} - 1\right)\right)^2 + \frac{2\gamma}{r_{gasket}}\right)\pi r^2_{reservoir} \qquad \textit{Equation 9}$$

Using the fluids velocity in the reservoir (assumed to be 0.025 m/s given the programmed motor speed) and the standard properties of water, such as surface tension and density [25], the graph depicted in Figure 12 was plotted using Desmos [26]. The graph represents the relationship between the gasket hole radius and the stepper motor force necessary for fluid transfer to occur, with a range (constrained by the dimensional and mechanical feasible available in the system) of lines corresponding to reservoir diameters (full flow model derivation in Appendix 1).

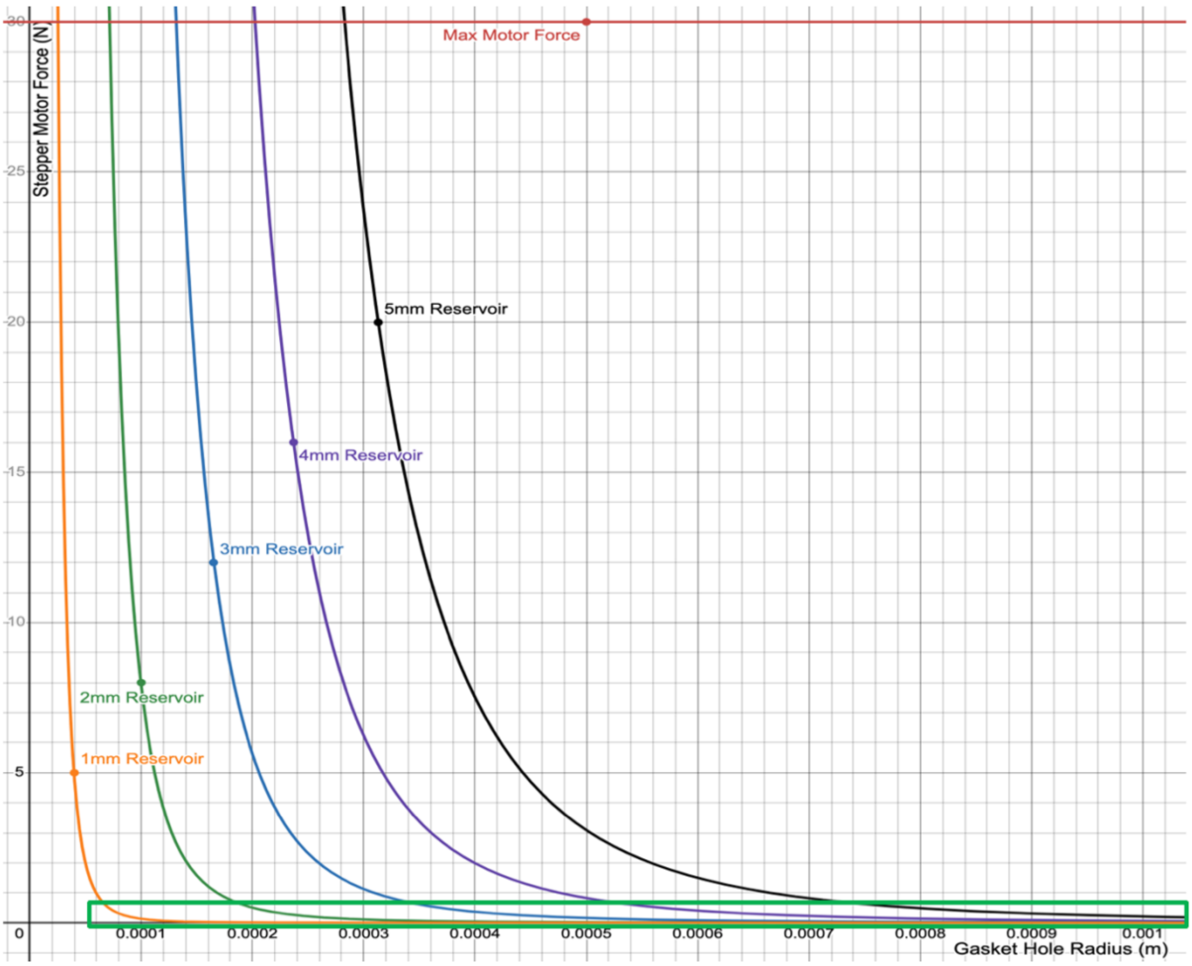

Figure 12: Gasket Hole Radius against Stepper Motor Force for a Range of Reservoir Hole Radii.

The optimal region of interest is highlighted by a green box, where the motor force remains below 1 Newton. Within this designated area, changes in hole size have a negligible effect on the required motor force, thus satisfying the fluidic system's third requirement of "utilising minimal energy for fluid transfer". This zone suggests an array of optimal reservoir and gasket hole sizes that would minimise the motor force during fluid transfer. Nevertheless, the potential for these sizes to provoke leakage is a factor that must be tested experimentally.

An additional model named the Hydraulic Model (in Appendix 2) was used to validate the Flow Models findings. It suggested lower force requirements than Flow Model. The Flow Model has been preferred for setting the range of hole sizes to veer on the side of caution, accounting for higher forces. Prior to proceeding with physical experiments for the hole sizes within the range chosen, the potential deformation of an elastomeric gasket under pressure, which could alter the gasket hole dimensions and hence the determined optimal range, will be considered in the following section.

3.2.3. GASKET MODEL

The Gasket Model FEA within ABAQUS is a necessary element to the theoretical analysis given the hyperelastic properties of the silicon rubber gasket. The model is used to give the relationship between gasket pressure and gasket hole size. The model will assess how the gasket deforms and experiences stresses and strains under varying loading conditions, while determining what pressures allow for the gasket to remain intact and not rupture or deform excessively.

To reduce the processing time and complexity of the FEA model, the geometry was simplified from the full SU's 4 holes to a single hole within a gasket. This was feasible since the four holes are all entirely independent of one another given no connecting airways or fluid flow paths. The simplified model can be seen in figure 13 below:

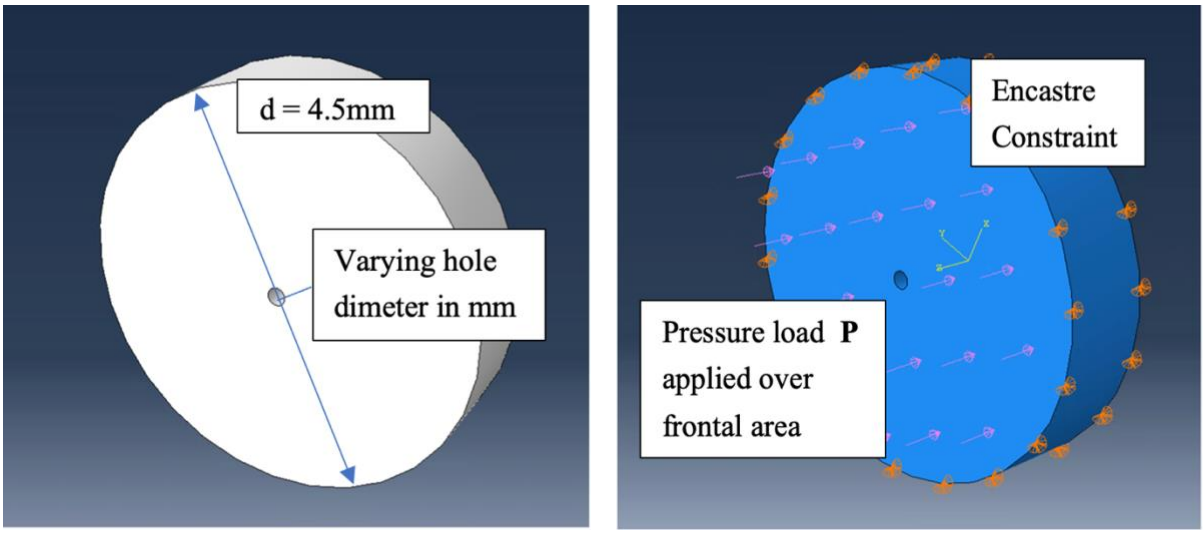

Figure 13: Left: Simplified Model for Gasket Analysis. Right: Model Constraints and Load.

Figure 13 shows the model constrained and loaded in Abaqus. The table below describes the model parameters.

Table 2: FEA Model Parameters and Explanations.

Parameter	**Explanation**
GEOMETRY	A 3D continuum geometry is used as opposed to a 2D planar stress geometry given that the latter would not accurately approximate the problem due to the inhomogeneous stress distribution throughout the z-axis.
MATERIAL PROPERTIES	The material used is silicone rubber with hyperelastic pre-sets defined using generalised Ogden model parameters sourced from the published 2022 paper [27]. While this is sufficient for preliminary testing, precise characteristics of the stress-strain relationship of the material will be obtained for critical tests either through the manufacturers data or through physical tests conducted in the lab.
ANALYSIS TYPE	A static analysis was used as the pressure is not varied over time and is assumed constant.
DISPLACEMENT BOUNDARY CONDITIONS	The geometry is restrained around the circumference of the disc using an Encastre boundary conditions limiting displacement and rotation in all x, y and z directions.
APPLIED LOADS	A pressure load is applied to the surface face as seen in the figure above. The magnitude of the load is varied for each pressure test.
ELEMENT TYPE	The element type used is A20-node quadratic brick, hybrid, linear pressure, reduced integration elements. Quadratic and hybrid were deemed appropriate, respectively, due to the curvature of the geometry and hybrid given the hyperelastic nature of the material.
ELEMENT SIZE	An element size of 0.2 mm is used for the outer edges of the gasket and 0.01 mm around the hole to allow for greater accuracy of results since this is the main region of interest.

The following assumptions were made in the creation of the model:

1. Material used in the model is homogeneous and isotropic.
2. The silicone rubber and fluid are both incompressible.
3. The temperature will remain constant.
4. Viscosity of the fluid will remain constant given constant temperature.
5. The pressure is applied uniformly to the gasket

The displacement around the hole was examined in the FEA model under varying pressure loads. The assessment involved measuring the displacement using the distance tool in ABAQUS on the points of maximum z axis displacement. An example displacement figure can be seen below highlighting the blue region of interest for a hole diameter of 2 mm and an applied gasket pressure of 3MPa.

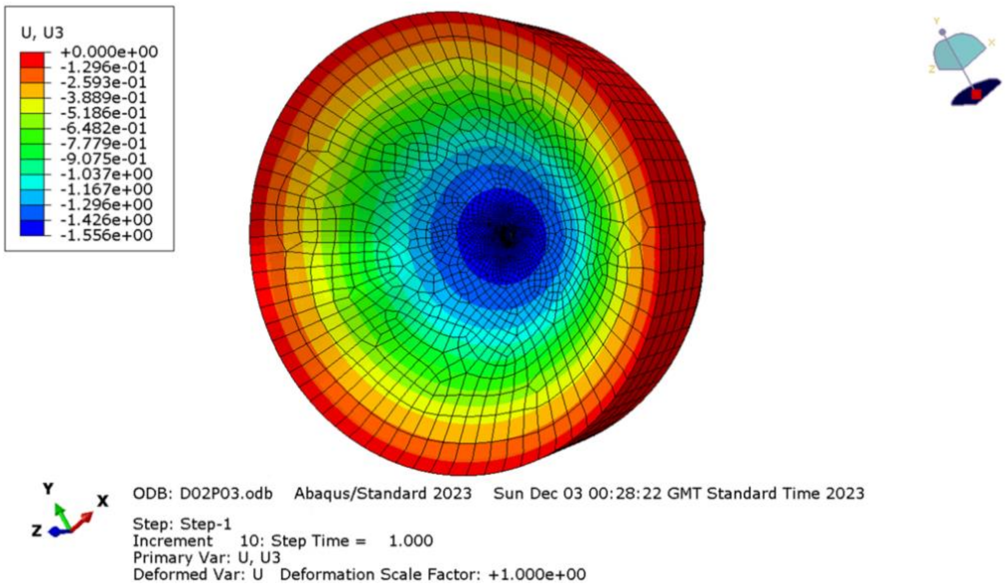

Figure 14: FEA Model of displacement in z axis (U3) with hole diameter of 2 mm and applied gasket pressure of 3MPa

Establishing the relationship between the applied pressure on the gasket change in hole size required several simulations with varying applied pressures from 0.01 to 3 MPa, and hole diameters from 1 to 3 mm. The results for each change in diameter were averaged out and plotted against their respective applied pressures seen in the graph below.

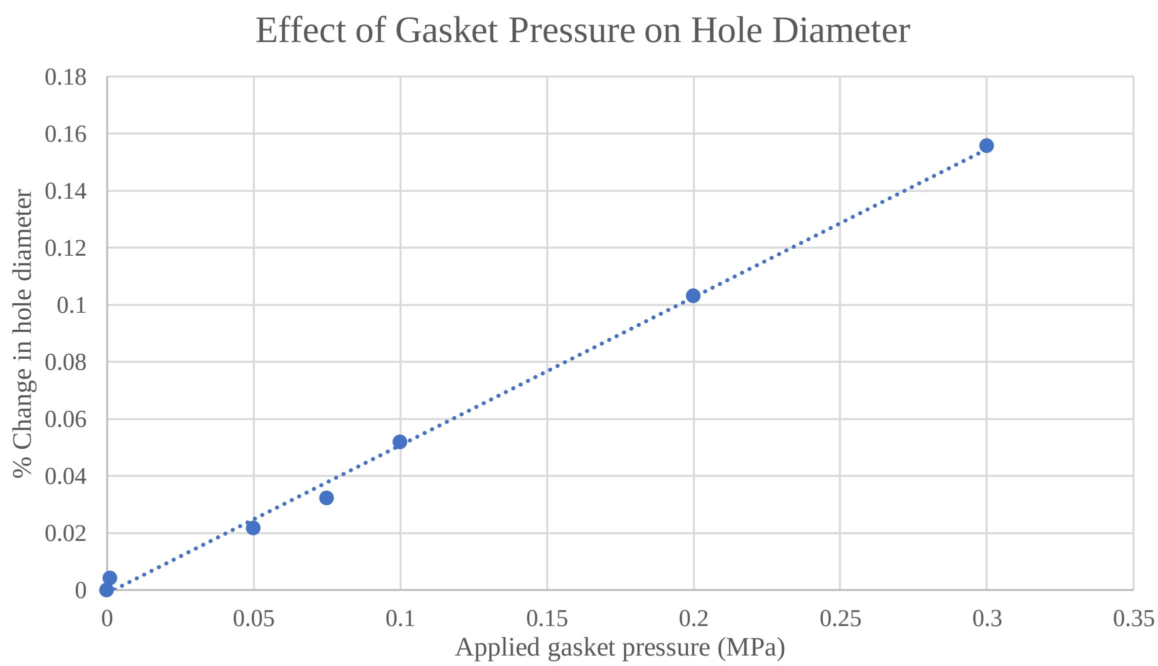

Figure 15: Graph of Gasket Pressure against Change in Hole Diameter (full results in Appendix 3).

The results showed the correlation between applied pressure on the gasket and the percentage increase in the hole diameter, which can be represented as a linear line of best fit. This relationship, applicable to any hole size, gives an equation that can be rendered down to the form seen below where "d" represents the initial gasket hole diameter and "P" denotes the applied gasket pressure in MPa.

$$\%\Delta d = 0.519P \qquad \textit{Equation 10}$$

When considering the changing gasket hole size with applied fluid pressure as modelling in FEA, the results showed the gasket hole would increase when pressure was applied which in turn would decrease the force exerted by the motor. Nonetheless, the increase in hole size due to pressure denoted by the equation is seen to be minor and therefore negligible for the theoretical model at hand, especially considering the relatively small forces, and therefore pressures, established by the Flow Model optimal hole range.

3.2.4. PLUNGER FRICTION MODEL

Given the relatively small forces predicted by the theoretical model when compared to the motor's maximum force output, an additional friction model was developed to estimate the frictional effects between the plunger and the reservoir walls. The slow fluid flow doesn't allow for the use of Reynolds number, and consequently, the Darcy friction factor is not applicable for determining the friction force between the plunger and the reservoir walls.

Instead, an alternative method can be used (as illustrated in the figure below) which utilises the compression of the rubber plunger heads as an indicator of the normal force exerted on the reservoir walls which is then multiplied by the dynamic friction coefficient between the rubber and aluminum surfaces to provide an approximate measure of the frictional force.

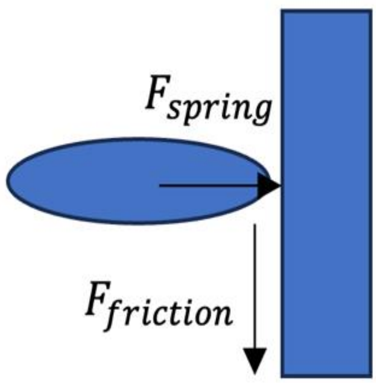

Figure 16: Friction Model Diagram

This frictional force represents the energy lost to friction as the plunger moves within the reservoir, described by the following equation:

[redacted]

$$F_{friction} = F_{spring} \times \mu_{dynamic} \qquad \textit{Equation 11}$$

Finding the frictional force firstly requires the calculation of the spring or normal force through the spring force equation (where x is the change in the rubber size):

$$F_{spring} = kx \qquad \textit{Equation 12}$$

The values of for the spring coefficient of silicon rubber [28] and change in the O-ring diameter (1% change) were substituted into the equation as follows:

$$F_{spring} = 2.69 \times 10^6 \times 0.000046 = 123.74\ N \qquad \textit{Equation 13}$$

Substituting the newly found spring force into the original friction equation 11 as well as the friction coefficient between aluminium and rubber [29] gives:

$$F_{friction} = 0.76 \times 123.74 = 94.04\ N \qquad \textit{Equation 14}$$

This result, although far higher than the maximum motor force, serves as an indicator toward the larger impact friction has in the system compared to hole size changes. Therefore, the friction effects must be a priority when considering the final design.

3.3. EXPERIMENTAL VALIDATION

3.3.1. PLUNGER O-RING TESTING PROCEDURE

The Plunger Friction Model's findings showed that the friction between the plungers and reservoir walls was the primary contributor to the force exerted on the motor. To address this, the plunger design was revised from the initial syringe tip configuration to an O-ring design as seen in figure 16 below:

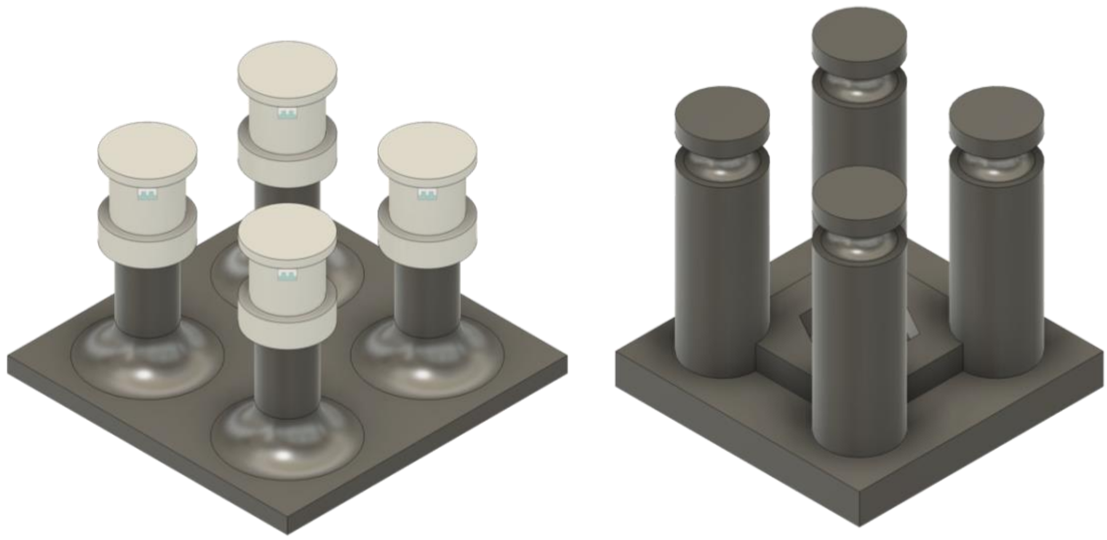

Figure 16: Left: Previous Syringe Tip Plunger Design. Right: New O-ring Plunger Design.

[redacted]

This modification aimed to reduce the contact surface area between the plunger and the reservoir walls, thereby diminishing frictional forces and therefore motor energy usage. To determine the effectiveness of the newly proposed O-ring design against the previous syringe-tip model, an experiment was set up. The objective was to measure the force exerted by friction between the two plunger designs and the reservoir walls and ascertain if the new design reduces this frictional force, thereby improving efficiency.

For the experiment, the necessary components, including both the original and new plunger plates as well as the test base, were fabricated using 3D printing. Each plunger plate was then affixed to the reservoir layer. The assembled setup was placed on an electric scale, which was zeroed before initiating the test to ensure accuracy. Mass was progressively added to the reservoir layer, applying a load that prompted the plungers to descend into the reservoir wells. The scale's measurement, when the plunger overcame the frictional resistance and descended fully, was recorded. The mass observed in grams was then converted to weight in N, by multiplying it by gravitational acceleration to calculate the friction-induced force. To ensure the reliability of the data, this process was repeated three times for each plunger design.

Figure 17: Plunger Testing Setup, 1: Mass, 2: Reservoir Layer, 3: Plunger Layer, 4: Scale.

3.3.2. HOLE SIZE TESTING

The theoretical Flow Model predicted an optimal range of hole sizes for both the reservoir and gasket within the fluidic system, which required experimental verification to confirm the practical viability of these dimensions.

Thus tests were conducted for this purpose using a series of gasket layers with holes varying from 1.04 mm to a maximum of 2 mm in diameter as seen in figure 18, all within the optimal range with a limit of 2 mm, being the size of a standard water droplet [30]. The process of creating the gasket holes was relatively simple, involving the puncturing of rubber squares with needles of specified sizes. In contrast, manufacturing the aluminium reservoirs was a more complex procedure that included milling and drilling, resulting in a comparatively limited set of reservoir sizes available for testing.

An important preliminary stage of the testing involved assessing the clearance between the plunger size and the reservoir hole. This commenced with an existing 4.6 mm diameter reservoir, around which three gasket hole sizes of 4.5, 4.6, and 4.7 mm were trialled to determine the optimal fit for a 4.4 mm diameter plunger. The clearance that proved most effective would inform the dimensions for future reservoir iterations. Corresponding O-ring plungers were 3D printed for the 3 sizes, and the testing apparatus was constructed using a linear actuator motor, a Raspberry Pi for control and a power source.

Water was used to simulate the rehydration solution given its similar physical properties to the rehydration solution. The linear actuator motor was activated using the rasperberry pi and pushed the 40 uL of water from the wells through the gasket layer until the plunger made contact with the gasket layer. Any fluid transfer and leakage was observed and recorded and the plunger motion was reversed pulling the water back from the sample region to the reservoirs. This method was repeated with each gasket layer and then the entire procedure was repeated with each well. This approach would provide results which could be directly compared to the theoretical model, and validate the design assumptions.

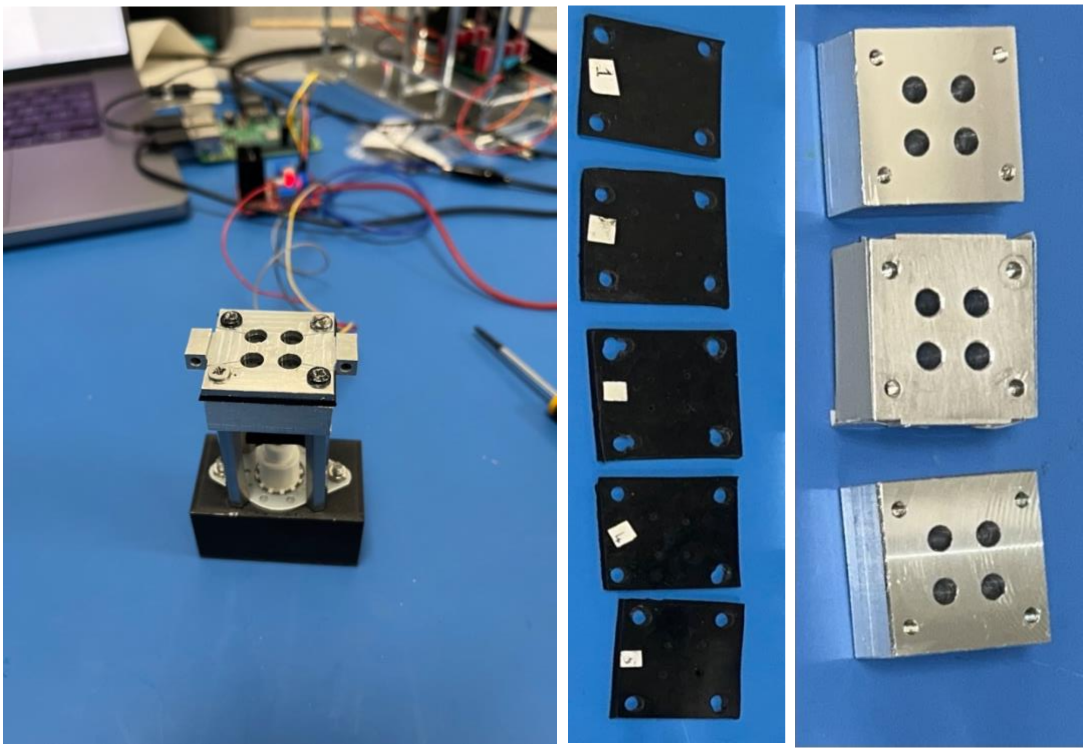

Figure 18: Hole Size Testing Setup Right: Assembled SU connected to Raspberry Pi, Middle: Gasket Layers with a range of punched hole sizes (dimeters: 1.04, 1.27, 1.6, 1.8, 2 mm), Right: Reservoir Layers (diameters: 4.5, 4.6, 4.7 mm)

3.3.3. ENVIRONMENTAL TESTING

Environmental testing is required to evaluate the integrity of the system under conditions mimicking launch vibrations. Vibration testing, performed on a shaker table, will subject the prototype with the selected gasket and reservoir hole sizes (determined in the prior tests) to the frequencies similar to those expected during launch. This will ultimately functionally validate the design against potential leakage issues encountered.

A prior VITA FEA was utilised as a data source for the first six natural frequencies of the spacecraft's shell [3]. These frequencies, detailed in Table 3 will inform the frequency settings for the shaker table.

Table 3: FEA results of first six natural frequencies identified.

Mode	**Frequency [Hz]**
1	1255
2	1586
3	1715
4	1778
5	2077
6	2552

4. RESULTS, DATA ANALYSIS AND DISCUSSION

4.1. PLUNGER O-RING TESTING

The plunger O-Ring testing focused on assessing the effectiveness of a new plunger design intended to reduce friction and thus motor force requirements. The testing procedure compared the force needed to displace the original syringe-tip plunger design with the updated O-ring design.

The results indicated a substantial 71% reduction in the force required by the motor to operate the new O-ring plunger design compared to the original syringe-tip plunger. The table below summarizes the measured mass change and the corresponding force in Newtons, calculated by multiplying the mass (kg) by the gravitational acceleration (9.81 m/s²):

Table 4: Plunger O-Ring Test Results.

Design	**Average Measured Mass change (kg)**	**Force (N)**
Syringe Tip	3.98	39.0
O-ring	1.15	11.3

These findings demonstrate that the O-ring plunger requires considerably less force to achieve the same vertical displacement through the reservoirs, suggesting that the new design substantially reduces frictional resistance. This decrease in necessary force translates directly into enhanced efficiency and reduced power usage by the motor.

4.2. PRELIMINARY MODEL TESTING

Empirical testing of the fluidic system's functionality with various gasket and reservoir hole sizes (detailed fully in appendix 4) was performed to establish an optimal configuration that would meet the criteria for efficient fluid transfer without leakage. To pass the experimental validation, the system needed to fulfil the following criteria:

- The system had to ensure a majority transfer of the 40 µL of water from the reservoirs to the wells.
- There could be no leakage outside the SU.
- Water could not leak prematurely from the reservoirs to the sample wells without plunger force application.
- A partial return of the water to the reservoirs by reverse plunging was expected.

The test results indicated that the gasket holes with 1.8 and 2 mm diameters were more susceptible to side leakage and inconsistent in water delivery to the sample wells, with certain wells receiving full volumes while others failed to receive any, as seen in figure 19. The 1.04, 1.27 and 1.6 mm diameter holes demonstrated similar issues, albeit to a lesser extent. The most effective clearance for the 4.4 mm plunger was established at 1 mm with the 4.5 mm reservoir hole, as larger diameter well setups showed signs of leakage at the base of the reservoir near the plungers. These findings would inform the decision to apply the 1 mm tolerance to a larger 6 mm reservoir paired with a 5.9 mm plunger. This reservoir size was chosen as it falls within the optimal range. More could not be made given time limitations, however this test along, with the 4.5 mm reservoir diameter, will adequately verify the ranges validity provided it functions as intended.

Several problematic observations were documented throughout all of the testing iterations. These included substantial leakage from the gasket layers sides and the reservoir wells base, shown in figure 19. Such occurrences were sporadic across test repetitions and were attributed to multiple factors:

- The machining finish of the reservoir layer surface lacked the necessary smoothness, allowing for fluid leakage through tiny surface microchannels.
- Discrepancies in the timing of plunger entry into the reservoir holes suggested an uneven force distribution from the motor shaft on the plunger layer.
- Misaligned standoffs caused by inconsistencies in the threading, resulted in the plungers bending and not entering the reservoir holes correctly, allowing for leakage through this region.
- Variations in screw sizes (due to difficulty in acquiring the precise size necessary for the system) likely caused a discrepancy in the force applied to each attachment point of the gasket layer, leading to varying hole sizes and erratic outcomes.

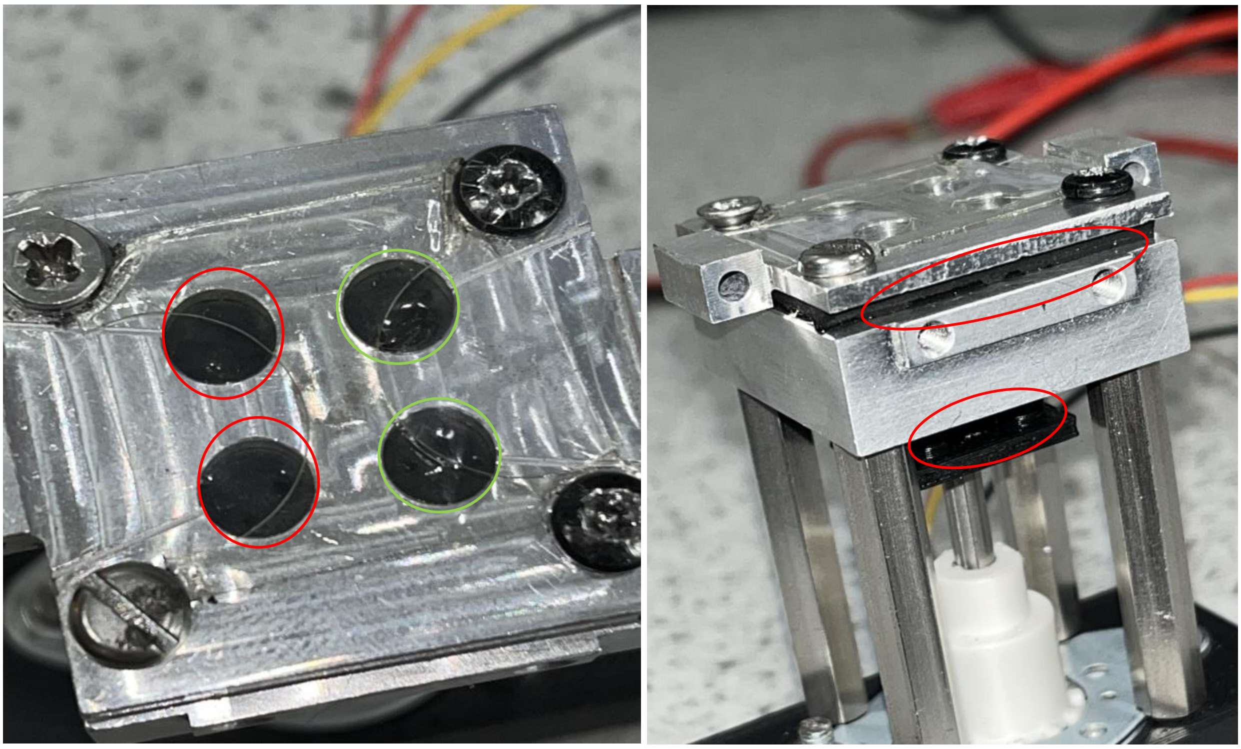

Figure 19: Left: SU system showing adequate fluid transfer (green) through only 2 of the 4 holes, Right: Leakage from the SU through the gasket sides and reservoir bottom onto the plunger layer.

To address these issues, several adjustments to the system were implemented. The reservoir layer was re-machined to ensure a smoother finish, thereby eliminating potential leakage microchannels. Reinforcement of the plunger base was undertaken by redesigning for a thicker base to provide the necessary support and prevent bending, given the uneven standoffs. The elastomeric gasket layer was to be heat pressed, to allow for tighter sealing of the layers, offering a sealed integration of the rubber gasket with the reservoir layer, thereby saving manufacturing time and costs. Moreover, the acquisition of the correct 7 mm screws resolved the force disparity issues across the gasket layer.

4.3. CRITICAL MODEL TESTING

The critical model with the applied changes was tested (detailed fully in Appendix 5), utilising the shortened range of gasket holes from 1.04 to 1.6 mm, reservoirs with 4.5 mm and 6 mm diameters and their respective newly 3D printed and reinforced plunger layers sized at 4.4 mm and 5.9 mm. The design choices stemmed from the previous conclusions drawn in the preliminary results section.

Heat seal in these tests was not resorted to, due to the operational complexity in removing the sealed gaskets after every test as well as potential material waste associated with its application. Given this omission, leakage in the gasket layer was still observed, however far lesser than before (as per the comparison in figure 20).

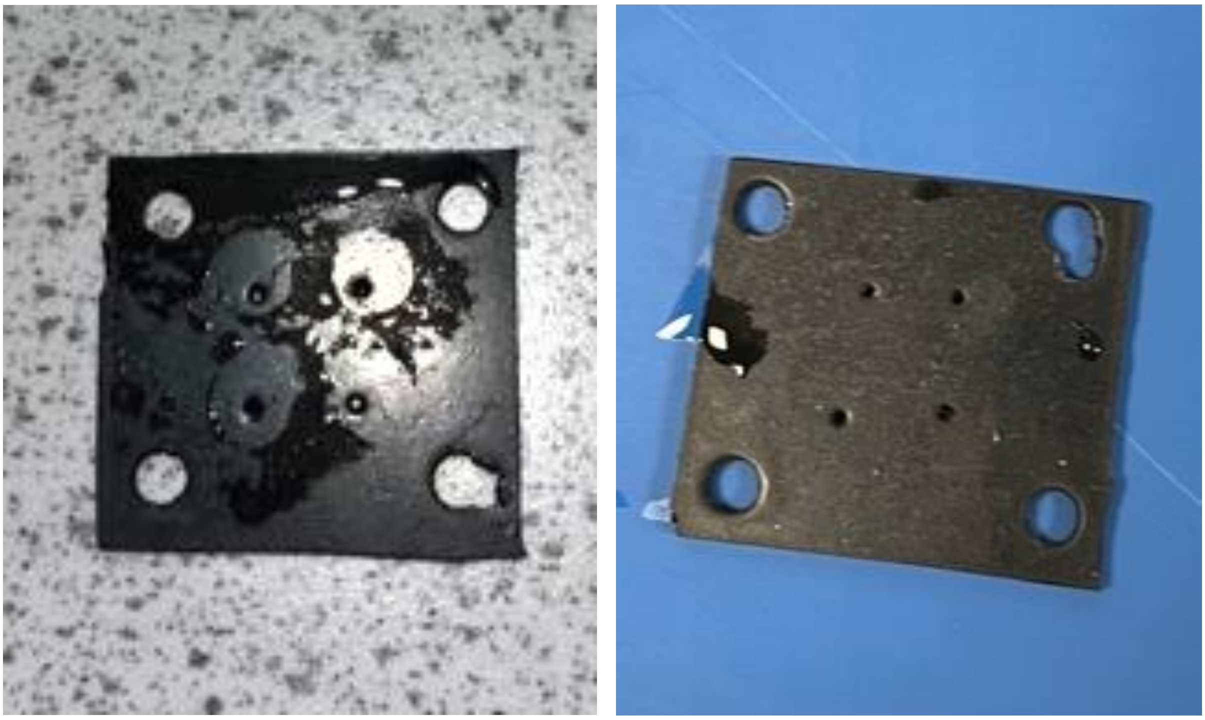

Figure 20: Left: Gasket layer with leakage before design modifications, Right: Gasket layer with less leakage after design modifications.

The results showed that both 4.5 mm and 6 mm reservoir configurations successfully exhibited optimal fluid transfer across all gasket hole sizes. However, the 4.5 mm reservoir model consistently produced less side leakage, supporting the Flow Model theory that a smaller reservoir diameter effectively reduces leakage through the plunger base. Furthermore, a smaller reservoir size would lessen the area for frictional forces between the plunger and the reservoir to occur. In turn this would decrease the force and therefore the energy demanded of the motor, giving further credence to the smaller reservoir hole size as the optimal design choice. Among the tested gasket layers, the 1.27 mm variant was distinguished as the most optimal choice, demonstrating minimised leakage whilst performing optimal fluid transfer. Tests on the optimal reseroir and gasket hole sizes were repeated to assure validity. The optimal configuration performance can be seen in the figure below.

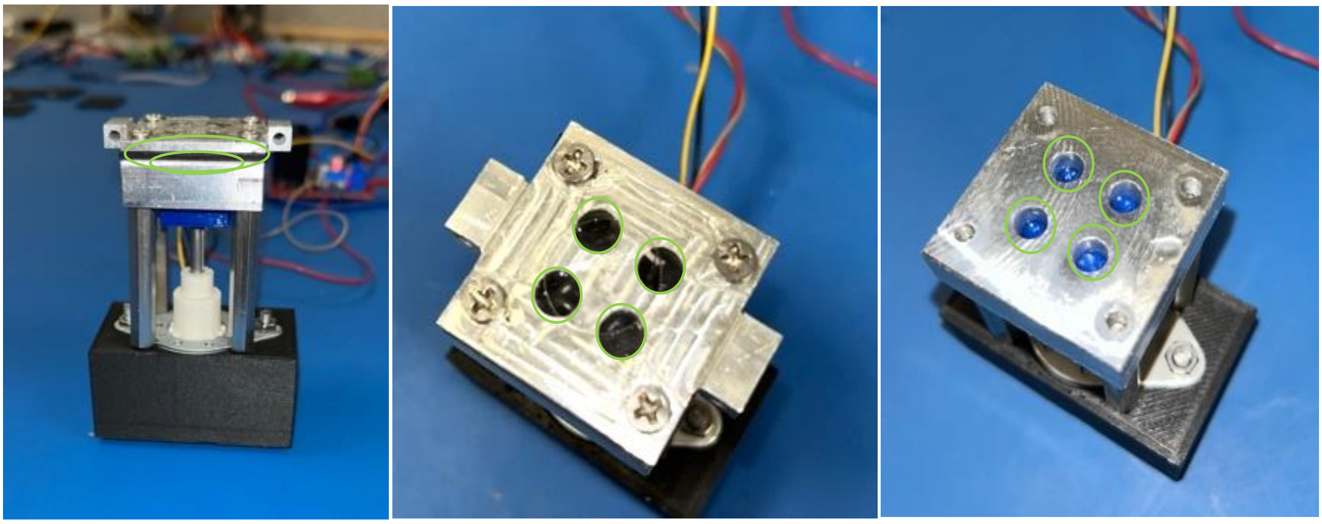

Figure 21: Optimal Hole Size Configuration 1.27 mm Gasket, 4.5 mm Reservoir, Left: Evidence of no side leakage from gasket sides or plunger layer interface, Middle: Fluid transfer through all 4 holes, Right: Partial reverse fluid transfer.

The chosen optimal configuration complies with the VITA requirements, ensuring the secure containment of the solution and preventing any unwanted leakage into the sample areas or outside the allocated reservoir boundaries. It will efficiently transfer 40 µL of the S30A buffer rehydration solution to the designated sample wells utilising the minimum energy necessary for effective fluid transfer, thereby adhering to the efficiency requirements. The heat seal method (when applied) will not employ use of chemical adhesives, thus eliminating the risk of contaminating the cell extracts and ensuring the integrity of the experiment. The location of the pre-punched gasket holes is aligned with air evacuation channels, ensuring that the samples can be adequately rehydrated under microgravity conditions. Reservoir holes have been chosen to fit within the confined 18 x 18 mm envelope. Overall the configuration meets the set criteria and will proceed to environmental testing alongside the remaining VITA system components upon their completion.

5. CONCLUSIONS

This project focused on optimising and validating the performance of an elastomeric gasket-based fluidic system, a new approach in space systems. Theoretical models were developed to approximate the behaviour of the fluidic system different reservoir and gasket hole size configurations. The models

[redacted]

indicated the absence of a perfect hole size for maximum efficiency but rather a realistic range of potential options that would minimise force required for fluid transfer within the constraints. Additionally, said models suggested that the main stepper motor loads arise from frictional forces between the plunger and reservoir walls.

To minimise the frictional forces a new O-ring plunger design was manufactured and tested, which successfully reduced frictional forces by 71%. Next the preliminary fluidic system designs featuring various hole sizes for the gasket and reservoir were physically prototyped and tested. These tests identified the optimal clearance between the plunger and reservoir but also exposed certain systematic design issues which resulted in leakage. The issues were addressed by improving the smoothness of the reservoir layer and redesigning the plunger layer for increased rigidity.

Empirical testing of the critical design configurations determined that a 4.5 mm diameter reservoir hole coupled with a 1.27 mm diameter gasket hole was optimal for fluid transfer, corroborating the theoretical model's predictions about force and leakage minimisation. The design demonstrated the absence of external system leakage and optimal fluid transfer complying with the VITA experiment requirements.

6. FUTURE WORK

Future work will involve environmental tests of the entire VITA system under launch conditions utilising a shaker table to simulate launch condition vibrations. This assessment will determine the systems propensity for leakage and whether it meets the operational standards for deployment. Moreover, the aforementioned heat-sealing techniques will be incorporated into the fluidic system to further guarantee the system’s integrity. These extra measures aim to address potential failure modes and increase the system’s reliability. Additionally, possible failsafe designs for the fluidics system will be looked into. This could involve a redesign of the SU sample well layer to incorporate a reservoir channel to manage unintended leakage.

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APPENDIX 1 - FLOW MODEL FULL DERIVATION

In utilizing Bernoulli's equation, the plunger pressure ($P_{reservoir}$) can be linked to the gasket pressure (P_{gasket}), whilst simplifying the terms of potential energy on account of equal hole centre heights, to give:

$$P_{reservoir} + \frac{1}{2}\rho V_{reservoir}^2 = P_{gasket} + \frac{1}{2}\rho V_{gasket}^2 \qquad \textit{Equation 15}$$

This can be rearranged to give a pressure and velocity relationship.

$$P_{reservoir} - P_{gasket} = \frac{1}{2}\rho(V_{gasket}^2 - V_{reservoir}^2) \qquad \textit{Equation 16}$$

Now the continuity equation (conservation of mass) [30] can allow the relation of radius for both the reservoir and gasket hole to fluid flow velocity:

$$\rho A_{reservoir} V_{reservoir} = \rho A_{gasket} V_{gasket} \qquad \textit{Equation 17}$$

$$\pi r_{reservoir}^2 V_{reservoir} = \pi r_{gasket}^2 V_{gasket} \qquad \textit{Equation 18}$$

$$V_{reservoir} \frac{r_{reservoir}^2}{r_{gasket}^2} = V_{gasket} \qquad \textit{Equation 19}$$

The V_{gasket} term can be substituted out of the pressure and velocity relationship which was derived above.

$$P_{reservoir} - P_{gasket} = \frac{1}{2}\rho\left(\left(V_{reservoir} \frac{r_{reservoir}^2}{r_{gasket}^2}\right)^2 - V_{reservoir}^2\right) \qquad \textit{Equation 20}$$

$$P_{reservoir} - P_{gasket} = \frac{1}{2}\rho\left(V_{reservoir} \frac{r_{reservoir}^2}{r_{gasket}^2} - V_{reservoir}\right)^2 \qquad \textit{Equation 21}$$

$$P_{reservoir} - P_{gasket} = \frac{1}{2}\rho\left(V_{reservoir}\left(\frac{r_{reservoir}^2}{r_{gasket}^2} - 1\right)\right)^2 \qquad \textit{Equation 22}$$

This equation above shows the relationship between the two pressures and now the Fluid Transfer Model equation 4 can be substituted to give the pressure terms in terms of the fluid transfer threshold.

$$P_{reservoir} - \frac{2\gamma}{r_{gasket}} = \frac{1}{2}\rho\left(V_{reservoir}\left(\frac{r_{reservoir}^2}{r_{gasket}^2} - 1\right)\right)^2 \qquad \textit{Equation 23}$$

[redacted]

$$P_{reservoir} = \frac{1}{2}\rho\left(V_{reservoir}\left(\frac{r_{reservoir}^2}{r_{gasket}^2} - 1\right)\right)^2 + \frac{2\gamma}{r_{gasket}} \qquad \textit{Equation 24}$$

$$P_{reservoir}A_{reservoir} = \left(\frac{1}{2}\rho\left(V_{reservoir}\left(\frac{r_{reservoir}^2}{r_{gasket}^2} - 1\right)\right)^2 + \frac{2\gamma}{r_{gasket}}\right)A_{reservoir} \qquad \textit{Equation 25}$$

$$F_{motor} = \left(\frac{1}{2}\rho\left(V_{reservoir}\left(\frac{r_{reservoir}^2}{r_{gasket}^2} - 1\right)\right)^2 + \frac{2\gamma}{r_{gasket}}\right)\pi r_{reservoir}^2 \qquad \textit{Equation 26}$$

It can be assumed that the $V_{reservoir}$ is approximately 0.025 m/s. Since we are testing using water, we can use the surface Tension of water is 0.072 [] , density of water is 998 kg/m3 [].

$$F_{motor} = \left(499\left(0.025\left(\frac{r_{reservoir}^2}{{r_{gasket}}^2} - 1\right)\right)^2 + \frac{0.144}{r_{gasket}}\right)r_{reservoir}^2\pi \qquad \textit{Equation 27}$$

Using a range of appropriate reservoir diameter within the space and mechanically feasible limitations the plot made in the Flow Model section was made demonstrating the fluid transfer thresholds for different hole sizes and their associated forces required to reach the thresholds, as well as a max motor force line denoting the maximum force capable by the motor.

APPENDIX 2 - HYDRAULIC MODEL

As an alternative to bring further credence to the flow model, the hydraulic model based on hydraulic principles [31], was utilised to determine the relationship between the motor force (F_{motor}) and the gasket pressure (P_{gasket}) instead of the Bernoulli principles. A simple hydraulic model demonstrates that the following would be true:

$$P_{reservoir} = P_{gasket} \qquad \textit{Equation 28}$$

Then the $P_{reservoir}$ term can be substituted with the force from the motor (F_{motor}) over the applied reservoir area ($A_{reservoir}$).

$$\frac{F_{motor}}{A_{reservoir}} = P_{gasket} \qquad \textit{Equation 29}$$

The Fluidic Transfer Model equation 4 can be substituted in for P_{gasket} to give the equation below.

$$P_{gasket} = \frac{0.144}{r_{gasket}} \qquad \textit{Equation 30}$$

$$F_{motor} = \frac{0.144\pi r_{reservoir}^2}{r_{gasket}} \qquad \textit{Equation 31}$$

It can be assumed that the reservoir diameter is 0.0046 mm for the creation of an example graph.

$$F_{reservoir} = \frac{0.144\pi r_{reservoir}^2}{r_{gasket}} \qquad \text{Equation 32}$$

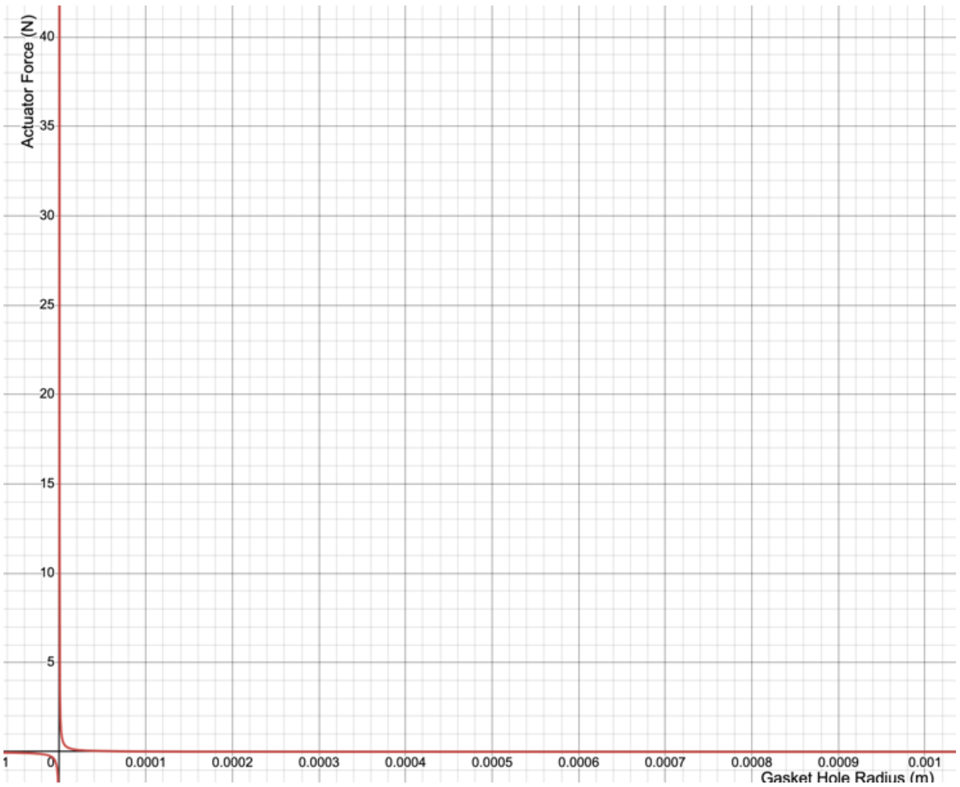


Figure 22 : Gasket Hole Radius against Stepper Motor Force for an example Reservoir Hole Radii.

APPENDIX 3 - FULL TABLE OF FEA RESULTS

Extended table of results used for FEA conclusions. Each table depicts a different pressure load applied on the gasket face and the corresponding hole diameter increases for each specific hole size.

Tables 5-10: Full results for each pressure iteration with respective hole size increases.

Pressure (MPa)	**0.3**
Hole diameter (mm)	Size Increase
1	0,156
2	0,312
3	0,465
Percent change of initial size	0,155666667

Pressure (MPa)	**0.2**
Hole diameter (mm)	Size Increase
1	0,107
2	0,209
3	0,294
Percent change of initial size	0,103166667

Pressure (MPa)	**0.1**

Pressure (MPa)	**0.075**

[redacted]

Hole diameter (mm)	Size Increase
1	0,0654
2	0,0911
3	0,134
Percent change of initial size	0,051872222

Hole diameter (mm)	Size Increase
1	0,0295
2	0,0702
3	0,0964
Percent change of initial size	0,032244444

Pressure (MPa)	**0.5**
Hole diameter (mm)	Size Increase
1	0,0233
2	0,0422
3	0,0612
Percent change of initial size	0,0216

Pressure (MPa)	**0.01**
Hole diameter (mm)	Size Increase
1	0,00426
2	0,00797
3	0,0118
Percent change of initial size	0,004059444

The values for percentage change of initial hole size were subsequently inserted into the table below which was used to produce the graph seen in the project progress section.

Tables 11: Applied gasket pressure against percentage change in hole diameter

Pressure (Mpa)	% Change in Hole Diameter
0.3	0,155666667
0.2	0,103166667
0.1	0,051872222
0,75	0,032244444
0,5	0,0216
0,01	0,004059444
0	0

APPENDIX 4 - FULL PRELIMINARY DESIGN TESTING RESULTS

Results from the full preliminary design testing are shown in the table below. The experiment started at 8 steps (approximately 4.8 mm) from the point at which the plunger would reach the gasket layer in

[redacted]

turn theoretically compressing all fluid. The notes from each experiment stage are separated into step intervals and include associated images for clarity.

<table>
<tr><td colspan="2" rowspan="2"></td><td colspan="3">O-ring Plunger Layer Diameter (mm)</td></tr>
<tr><td>4.5</td><td>4.6</td><td>4.7</td></tr>
<tr><td>Gasket Hole Layer Diameter (mm)</td><td>1.04 (3)</td><td>After 2 steps minimal fluid transfer occurred in one hole. After 3 steps more substantial fluid transfer was seen in 2 holes. After 4 steps fluid transfer was visible in all 4 holes.

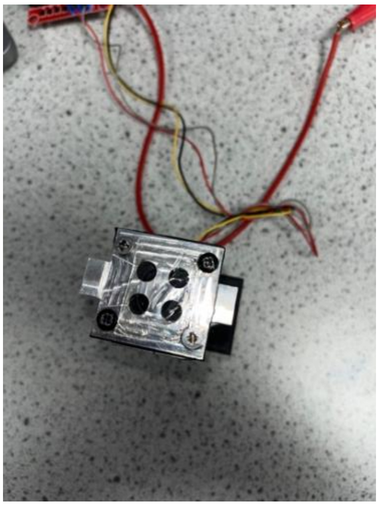
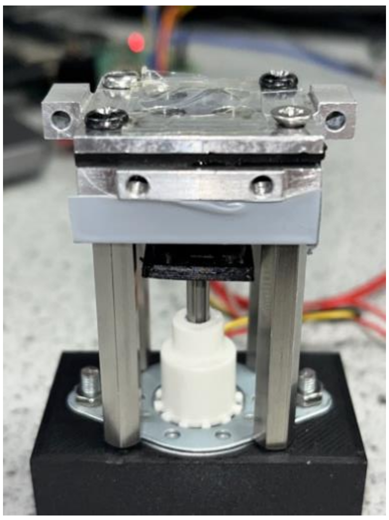



Figure 23: Gasket Fluid Transfer.

The trend continued until 8 steps where substantial fluid transfer occurred in 4 holes and little side leakage occurred. Reverse fluid transfer to reservoir occurred optimally where around half of the fluid was transferred back.
</td><td>After 4 steps minimal fluid transfer occurred in one hole. After 8 steps substantial fluid transfer occurred in one hole and partial fluid transfer occurred in 2 other holes. Substantial side leakage was prominent from two gasket sides as well from the plunger layer indicating an imbalance in the pushing force.

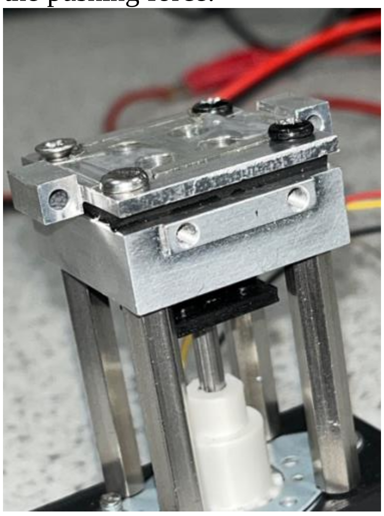



Figure 29: Plunger Layer Leakage.

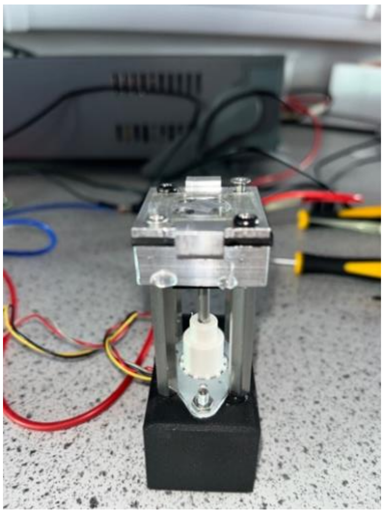



Figure 30: Gasket Side Leakage.

Substantial leakage from the plunger layer indicated the</td><td>Substantial water transfer in 1 hole after 4 steps however the other holes did not receive any. Plunger layer leakage is highly evident.

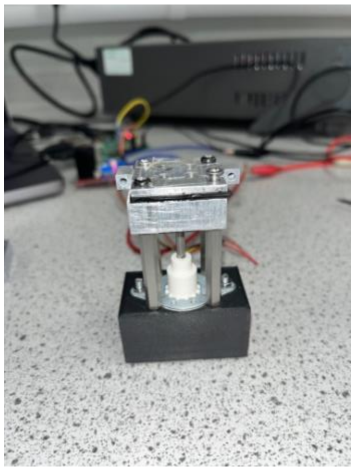
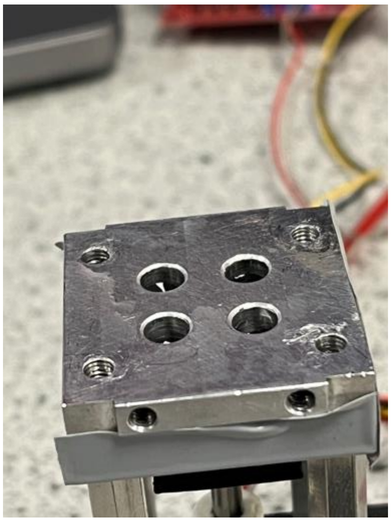



Figure 31: Gasket Side Leakage.

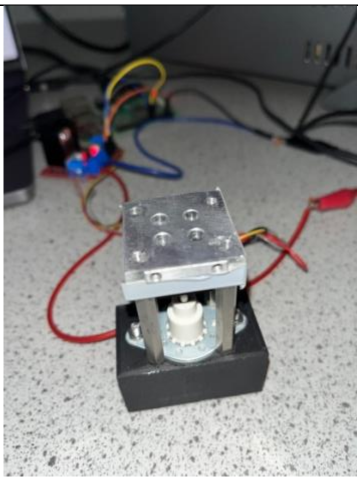
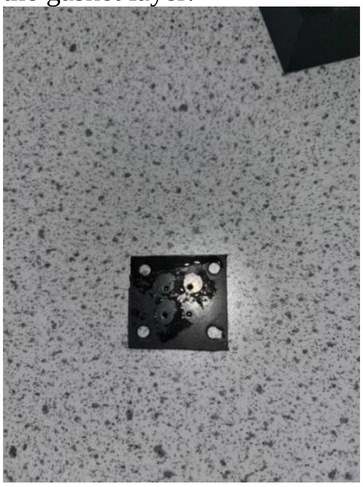
Figure 32: Plunger Layer Leakage.

Again, substantial leakage from the plunger occurred, demonstrating the optimal reservoir plunger layer tolerance to be the 1mm configuration.</td></tr>
</table>

		Figure 24: Gasket Side Leakge.	inadequacy of the plunger reservoir seal.	
	1.27 (4)	After 1 step side leakage from one gasket side occurred as well as no fluid transfer through any holes. After 4 steps there was fluid transfer in 3 holes. After 6 steps there was fluid transfer in the 4 holes and minimal side leakage from one side. However leakage from the gasket layer persisted.  Figure 25: Gasket Layer Leakage.  Figure 26: Gasket Side Leakage.	N/A	N/A

		 Figure 27: Successful Reverse Plunging.		
	1.6 (5)	After 5 steps fluid transfer occurred in 3 holes and minimally in one hole. After 6 steps substantial fluid transfer in 2 holes occurred. After 7 steps substantial fluid transfer in 3 holes occurred. More substantial fluid transfer after 8 steps in 3 holes leaving one without any suggesting an imbalance in force applied.  Figure 28: Gasket Fluid Transfer. Almost no reverse fluid transfer back to the reservoirs occurred.	N/A	N/A

[redacted]

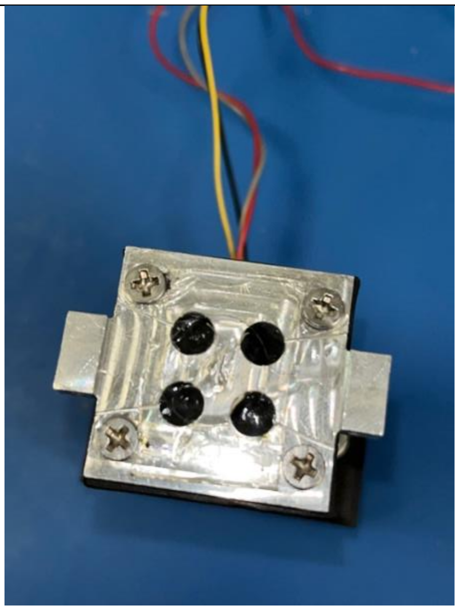
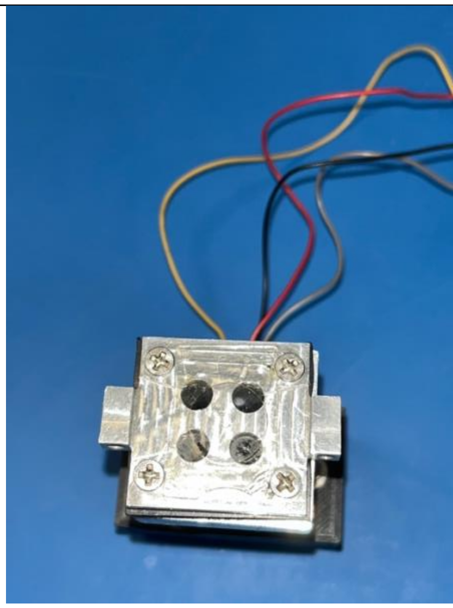
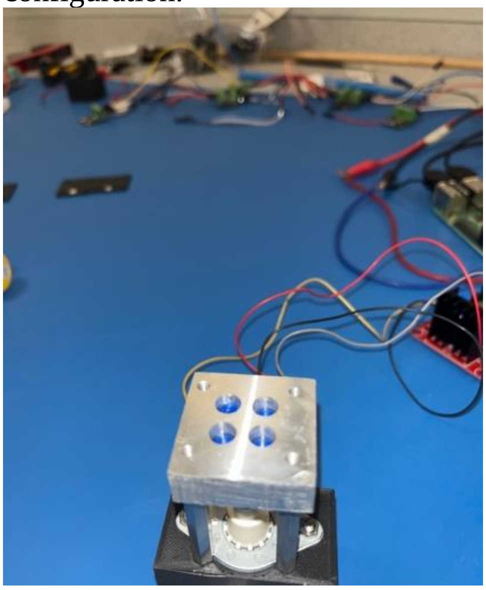
		 Figure 28: Unsuccessful Reverse Plunging. Substantial leakage occurred in the gasket layer.  Figure 29: Gasket Layer Leakage.		

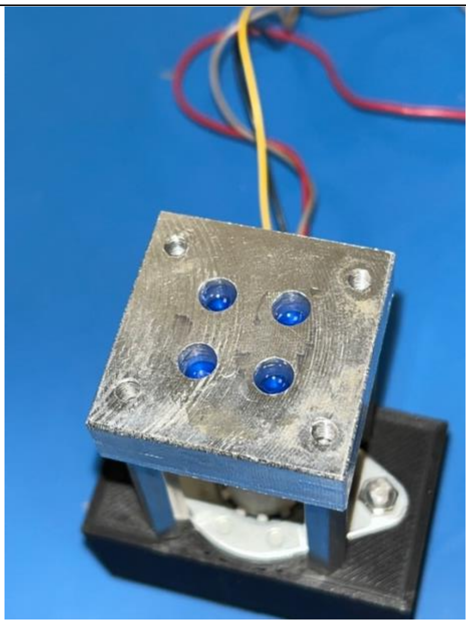
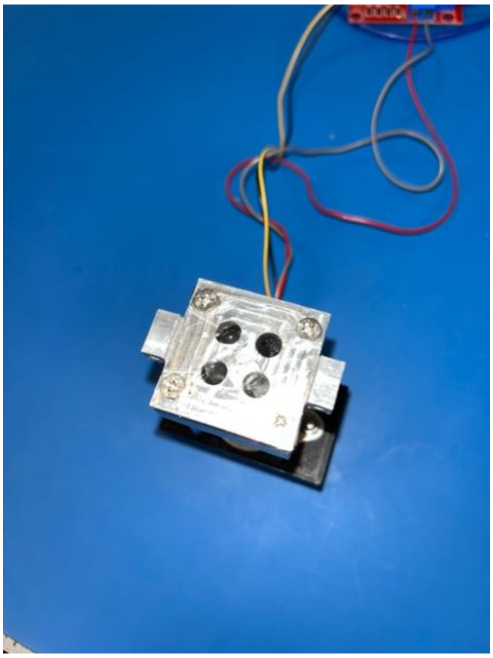
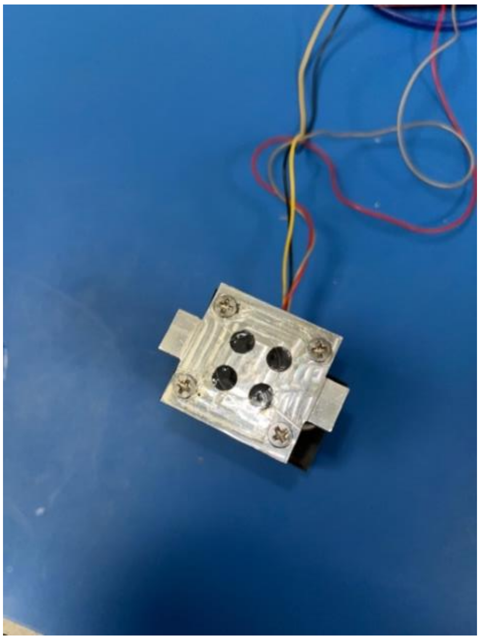
APPENDIX 5 - FULL CRITICAL DESIGN TESTING RESULTS

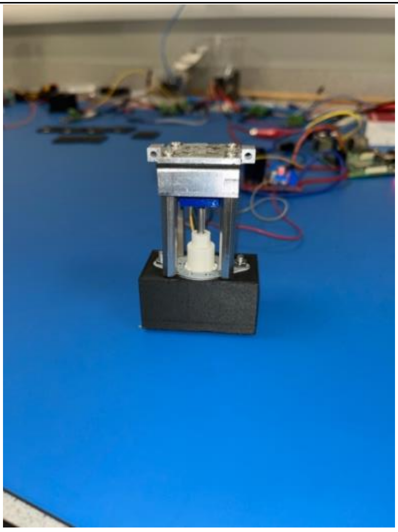
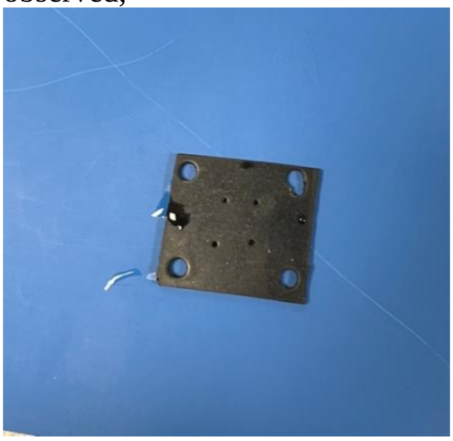
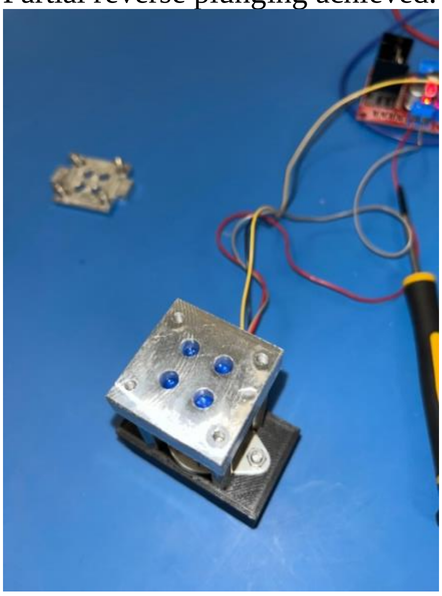
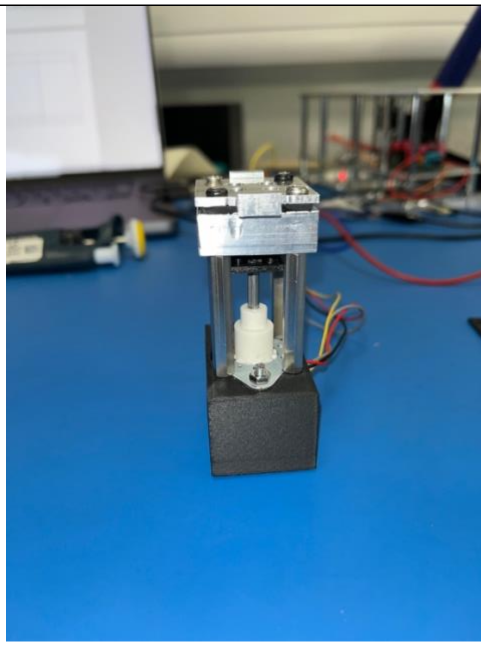
Results from the full critical design testing are shown in the table below. The experiment started at 6 steps (approximately 3.6 mm) from the point at which the plunger would reach the gasket layer. The notes from each experiment stage are separated into step intervals and include associated images for clarity.

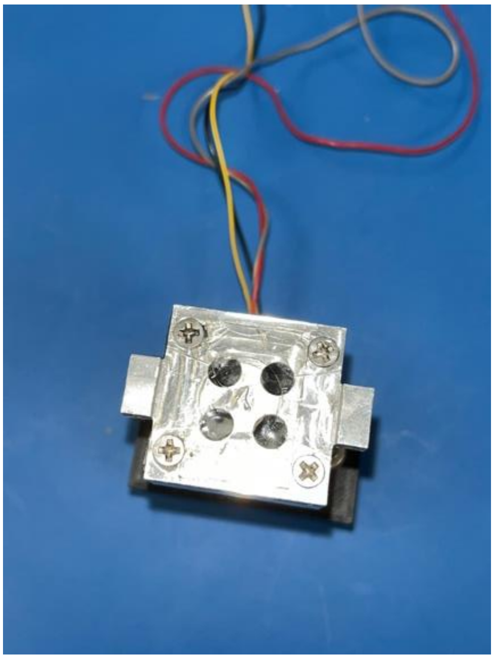
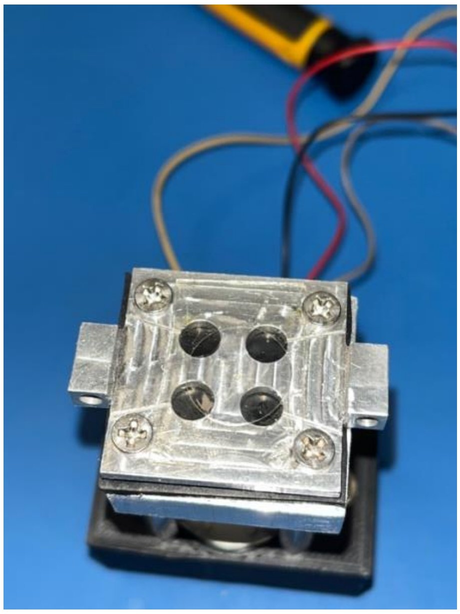
		4.5	6
Gasket Hole Layer Radius (mm)	1.03 (3)	After 2 steps, minimal fluid transfer occurred in one hole. After 4 steps, fluid transfer occurred in all 4 holes. After 6 steps, substantial fluid transfer occurred in all 4 holes.	After 2 steps no fluid transfer occurred. After 4 steps, fluid transfer occurred through 2 holes. After 6 steps, substantial fluid transfer occurred through 4 holes.

[redacted]

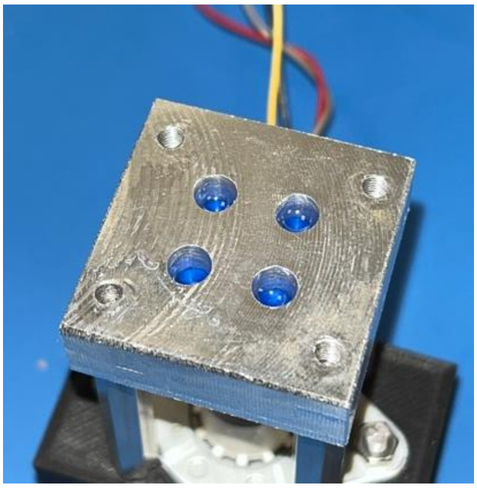
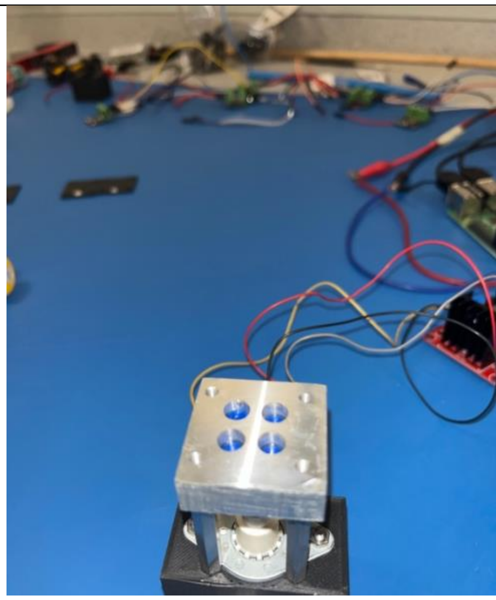
		 Figure 33: Optimal Gasket Fluid Transfer. Gasket layer leakage still occurred.  Figure 34: Gasket Layer Leakge. Partial reverse plunging was achieved.	 Figure 36: Optimal Gasket Fluid Transfer. Partial reverse plunging was achieved although less than in the 4.5mm configuration.  Figure 37: Partial Reverse Plunging. Slightly more gasket layer leakage was observed.

		 Figure 35: Optimal Reverse Plunging.	 Figure 38: Gasket Layer Leakage.
	1.27 (4)	After 2 steps, fluid transfer occurred in one hole After 4 steps, fluid transfer occurred in 3 holes. After 6 steps, fluid transfer occurred in all 4 holes.  Figure 39: Gasket Fluid Transfer. No side leakage was observed.	After 2 steps, fluid transfer occurred in one hole After 4 steps, fluid transfer occurred in 2 holes After 6 steps, fluid transfer occurred in 4 holes.  Figure 43: Gasket Fluid Transfer. Less gasket side leakage but still some.

		 Figure 40: No Gasket Side Leakage. Minimal gasket layer leakage observed,  Figure 41: Gasket Layer Leakage. Partial reverse plunging achieved.  Figure 42: Partial Reverse Plunging.	 Figure 44: Minimal Gasket Side Leakage. More gasket layer leakage was observed.  Figure 45: More substantial Gasket Layer Leakage.

	1.6 (5)	After 2 steps, fluid transfer occurred in 2 holes. After 4 steps , fluid transfer occurred in all 4 holes. After 6 steps, fluid transfer occurred in all 4 holes.  Figure 46: Gasket Fluid Transfer. There was slightly more gasket layer leakage observed.  Figure 47: Gasket Layer Leakage. Partial reverse plunging achieved.	After 2 steps, fluid transfer occurred in 2 holes. After 4 steps, fluid transfer occurred in 3 holes. After 6 steps, fluid transfer occurred in all 4 holes.  Figure 49: Gasket Fluid Transfer. Substantial gasket layer leakage was observed.  Figure 50: Substantial Gasket Layer Leakage. Partial reverse plunging achieved.

[redacted]

		 Figure 48: Partial Reverse Plunging.	 Figure 51: Partial Reverse Plunging.